Supply Chain Planning and Optimisation

Course 5 — Advanced Supply Chain and Strategic Logistics Management · Module 13 · Teaching content for the Theory Builder

Review copy. Each lesson lists its blocks in entry order: enter each block's heading and body into the Theory Builder; enter practice questions as Lesson Checks; enter the module assessment via Configure assessment. Grey appendix sections are audit notes, not for entry.

Lesson 13.1 — Demand Forecasting and Supply Planning

Summary

This lesson examines how organisations anticipate future customer demand and plan the resources required to meet it. Learners will explore forecasting methods, historical demand analysis, seasonal variations, supply planning, and the relationship between demand forecasts and procurement, inventory, production, and distribution decisions. The lesson also introduces forecast accuracy and the operational consequences of inaccurate predictions.

Learning objectives

- Define demand forecasting and supply planning.
- Explain the importance of forecasting in logistics and supply chain management.
- Identify qualitative and quantitative forecasting methods.
- Describe how historical sales data and seasonal patterns influence forecasts.
- Explain the relationship between demand forecasts and inventory requirements.
- Identify factors affecting supply availability and replenishment planning.
- Distinguish between forecast demand and actual demand.
- Calculate simple demand forecasts using historical data.
- Explain how forecast errors affect stock availability, transportation, and operating costs.
- Recommend basic measures for improving supply planning accuracy.

Key terms

Demand Forecasting — The process of estimating future demand for goods or services using available information and forecasting methods.

Supply Planning — The process of determining how resources, inventory, and supply activities will meet expected demand.

Historical Demand — Recorded customer demand from previous periods used to support forecasting and planning.

Qualitative Forecasting — Forecasting based on expert judgement, market knowledge, or other non-numerical information.

Quantitative Forecasting — Forecasting based on numerical data and statistical or mathematical methods.

Moving Average — A forecasting technique that calculates an average from a specified number of recent observations.

Seasonality — A recurring pattern of demand changes associated with particular periods or events.

Forecast Accuracy — A measure of how closely predicted demand matches actual demand.

Forecast Error — The difference between forecast demand and actual demand.

Demand Variability — Changes or fluctuations in customer demand over time.

Lead Time — The period between initiating a supply or replenishment activity and receiving the required goods.

Supply Capacity — The quantity of goods or services that a supply system can provide within a specified period.

Sales and Operations Planning (S&OP) — A coordinated business planning process that aligns demand, supply, and financial objectives.

Demand Planning — The activities involved in analysing and estimating customer demand to support operational decisions.

Teaching content

KEY CONCEPT

The planning layer: what demand forecasting and supply planning are

This lesson opens Module 13, Supply Chain Planning and Optimisation, and with it Course 5, the advanced level of this programme. The twelve modules before it taught the operational layer: how goods move, how they are stored and controlled, what the documents and procedures require, what logistics costs, and how risk and cross-border coordination are managed. This module moves one level up, to the layer that decides what those operations will be asked to do. Everything in operations consumes a plan, and every plan begins with an estimate of future demand.

Demand forecasting is the process of estimating future demand for goods or services using available information and forecasting methods. Supply planning is the process of determining how resources, inventory, and supply activities will meet expected demand. Around the first sits demand planning, the broader set of activities involved in analysing and estimating customer

demand to support operational decisions: gathering the data, choosing the method, reviewing the numbers with the people who know the market, and maintaining the forecast over time.

The two activities answer complementary questions. The forecast answers how much demand is expected, for what, where, and when. The supply plan answers what must be bought, made, moved, and held so that the organisation can serve that demand at an acceptable cost. Neither is ever perfect, and this lesson treats imperfection as a subject in its own right: forecast error is measured, budgeted for, and managed.

One habit distinguishes professional planners from beginners from the first block onwards. A forecast is not a wish, an order, or a promise. It is an input to decisions, and its quality is judged afterwards against what actually happened. The rest of this lesson builds that habit: first the methods, then the errors, then the plans that turn a forecast into purchasing, inventory, production, and distribution commitments.

KEY CONCEPT

Why forecasting governs every logistics decision

Forecasting matters in logistics because almost every logistics decision is made before demand happens. A decision made today about stock, space, vehicles, or purchase orders must be based on some assumption about what customers will want tomorrow. Consider how many decisions silently contain a demand number.

1. Inventory levels. The reorder point taught in Lesson 8.3, Inventory Control, FIFO, and FEFO, is built from expected demand during the lead time plus safety stock. Change the demand assumption and the correct reorder point changes with it.

2. Warehouse space. A warehouse sized for quiet months overflows in peak season, and a warehouse sized for peak season stands half empty and half paid for the rest of the year.

3. Transport capacity. Vehicle fleets, delivery schedules, and carrier contracts are set in advance; a forecast that misses by a third leaves either unsold capacity or undeliverable orders.

4. Procurement and production. Purchase orders, supplier contracts, and production schedules are commitments made against expected demand, often weeks or months before the goods are sold.

When the demand assumption is wrong, the error does not stay in the planning department. It propagates into the physical operation. Under-forecasting produces stockouts, lost sales, and emergency purchases shipped at premium cost. Over-forecasting produces excess stock that ties up working capital, occupies space, and, for perishable or fast-changing goods, becomes obsolete stock written off entirely. All of these land in the cost families built in Lesson 10.1, Components of Total Logistics Cost: more holding cost, more emergency transport, more obsolescence.

This is the importance of forecasting in one sentence: logistics executes demand assumptions, so the quality of those assumptions sets the ceiling on how well any warehouse, fleet, or budget can perform. A brilliant operation executing a poor forecast still delivers stockouts and surplus.

KEY CONCEPT

Historical demand, seasonality, and demand variability

The raw material of most forecasting is historical demand, the recorded customer demand from previous periods. Recorded sales are the usual proxy, with a caution the last note of this lesson returns to: sales record what was bought, not always what was demanded, because a stockout records zero sales for demand that walked to a competitor.

Drawn over time, history shows patterns, and reading them is a core demand-planning skill. Three matter most.

1. Trend: demand rising or falling persistently over time, as a product grows or declines.

2. Seasonality: a recurring pattern of demand changes associated with particular periods or events. Festive-season peaks, back-to-school demand, and harvest-time movements of agricultural produce all repeat each year, and a planner who knows the pattern plans the build-up weeks in advance.

3. Random variation: the leftover month-to-month wobble that follows no identifiable pattern. This wobble is demand variability, changes or fluctuations in customer demand over time, and it never disappears; it can only be buffered.

Forecasting also has a granularity and a horizon, and the two involve a trade-off every planner should be able to state. Forecasts can be made at item level, for an individual SKU and location, or aggregated, for a product family, a region, or a whole business. They can be made for the short term, weeks ahead, or the medium term, months ahead. Item-level, short-horizon forecasts are what daily operations consume, but individual items are noisy and error is proportionally large. Aggregated, medium-horizon forecasts are steadier and more accurate, but too coarse to drive tomorrow's picking. The professional answer is a hierarchy: plan by family for the medium term, where the numbers are stable, then break the family plan down to item level for the short term, where operations need it.

KEY CONCEPT

Qualitative forecasting: forecasting without history

Qualitative forecasting is forecasting based on expert judgement, market knowledge, or other non-numerical information. It is the method of necessity whenever numbers are missing: a new product with no sales history, a market about to be entered for the first time, a regulation or technology change that breaks the pattern of the past. Its main instruments are three.

1. Expert judgement and market intelligence. Salespeople, distributors, and market specialists hold structured knowledge that no database contains: which customers are expanding, what competitors are doing, which tenders are in the pipeline. Written down and combined systematically, this knowledge becomes a forecast.

2. Market surveys. A market survey is a structured collection of expectations from customers or the channel: retailers asked how much they expect to stock, buyers asked about purchase intentions. Surveys bring evidence rather than anecdote, but they must be designed carefully, because stated intentions exaggerate what people actually do.

3. Delphi-style expert panels. In a Delphi process, experts answer independently and anonymously, a coordinator aggregates the responses and shares the summary with the panel, and the experts revise their answers in further rounds. Anonymity prevents the loudest voice from carrying the room, and iteration nudges the panel toward a considered, converging estimate.

Qualitative forecasting is structured opinion. Its weaknesses are the weaknesses of human judgement, optimism, anchoring on recent events, and inconsistency between occasions, and it cannot be reproduced the way a calculation can. Used well, it is not a poor substitute for statistics but the only available method for genuinely new demand, and later in the lesson it reappears as a corrective to statistics through the sales and operations planning conversation.

KEY CONCEPT

Quantitative forecasting: methods that learn from data

Quantitative forecasting is forecasting based on numerical data and statistical or mathematical methods. It requires history, and it assumes the pattern found in that history continues into the future. Three methods cover the introductory territory, and this course treats them as tools to understand rather than formulas to memorise.

1. Moving average. A moving average calculates the average of a specified number of recent observations and uses it as the forecast for the next period. Forecast next month at the average of the last three months and the random wobble of any single month is smoothed away. The method is simple, transparent, and slow: by averaging old months with new, it always lags a trend.

2. Exponential smoothing. Exponential smoothing also averages history, but it weights recent data more heavily than older data, using a smoothing weight that the planner sets. A low weight produces a stable, sluggish forecast; a high weight follows the data closely, wobble and all. It is the workhorse of short-term item-level forecasting because it is cheap to run, easy to update, and responds to change faster than a moving average of the same length.

3. Regression, or causal forecasting. Regression links demand to driver variables: forecasting ice-drink demand from temperature and price, cement demand from construction activity, or demand from the calendar of promotions. Where a stable, believable driver exists, regression

explains demand instead of merely extending it, which is why it is the natural home of medium-term planning.

The choice is a matter of fit. Stable, data-rich items reward quantitative methods; new or historically erratic items do not, and judgement must carry them. Forecasting software and machine-learning models automate these same principles at scale; they do not repeal forecast error.

KEY CONCEPT

Forecast error and forecast accuracy: bias versus noise

Actual demand is what customers genuinely wanted in a period. Forecast demand is what the organisation predicted. The two are never identical, and the discipline of forecasting consists largely of managing the gap between them. Forecast error is the difference between forecast demand and actual demand, calculated period by period. Forecast accuracy is a measure of how closely predicted demand matches actual demand, summarising those errors, usually over several periods.

Two standard measures do the summarising. MAPE, the mean absolute percentage error, takes the absolute error of each period, divides it by that period's actual demand, expresses the result as a percentage, and averages the percentages across periods. MAD, the mean absolute deviation, averages the absolute errors in plain demand units. MAPE speaks the language of management, how wrong in percentage terms, but it becomes unreliable when actual demand is close to zero, because dividing by a tiny number produces an enormous percentage; MAD keeps errors in units that warehouse and transport planners can use directly.

Measuring error is only the beginning; the pattern of errors is where improvement comes from, and the essential distinction is between bias and noise.

1. Bias is systematic error: errors that cluster in the same direction, period after period. A forecasting method that has under-estimated demand for six consecutive months is not unlucky; it is structurally wrong, usually because it is missing a trend, a driver, or a market change. The remedy is to fix the forecast: add the missing driver, reweight the method, or hear what sales knows.

2. Noise is random variation: errors scattered in both directions around a sound centre. Noise cannot be fixed, because it is not an error of the method but the genuine variability of demand. The remedy is not a better formula but a buffer, which is where safety stock enters.

Confusing the two is expensive. Treating bias as noise leads planners to buy ever-larger safety stocks to compensate for a model that could simply have been corrected; treating noise as bias leads to endless method-tinkering against variation no formula can remove.

KEY CONCEPT

From forecast to inventory requirements

The clearest place to watch a forecast become a physical commitment is the inventory policy. Lesson 8.3 built the reorder point: the stock level that triggers replenishment, calculated as the expected demand during the lead time plus safety stock. Both ingredients trace back to the work of this lesson.

The lead time is the period between initiating a supply or replenishment activity and receiving the required goods. It is not merely transport time: it includes order processing, preparation or production, transit and, across borders, customs clearance. While waiting for replenishment, the business must cover demand out of stock, so the forecast of demand during the lead time sets the base of the reorder point.

Safety stock is the buffer held against the gap between expected and actual: it exists precisely because demand and supply both vary. Its size grows with the variability it must absorb, and with the service level targeted, the probability of avoiding a stockout; higher service targets require disproportionately more safety stock, a relationship developed in Lesson 13.3, Inventory Optimisation and Service Levels.

Two consequences follow, and they are the operational heart of this lesson.

1. Forecast error becomes stock policy. If the forecast used in the reorder point is accurate on average but scattered, the scatter is a known, budgetable quantity and safety stock absorbs it. If the forecast is biased, every replenishment cycle starts from the wrong base, and the buffer is perpetually compensating in one direction: recurring stockouts under a forecast that is persistently low, creeping surplus under one that is persistently high.

2. Lead time multiplies exposure. A forecast that must stay accurate for two days is rarely wrong by much; the same forecast stretched over a ten-week import lead time carries ten weeks of accumulated error. Long-lead items are therefore the forecast's most demanding customers, and they justify the planner's closest attention.

KEY CONCEPT

What supply planning must respect: capacity, lead time, and money

A demand plan, however sound, is only half the planning problem. Supply planning is the discipline of answering it, and a supply plan is only honest when it is built against the real constraints of supply. Supply capacity is the quantity of goods or services that a supply system, a supplier, a factory, a fleet, a warehouse, can provide within a specified period, and it is never infinite. The factors a planner checks before committing to a demand plan are the substance of replenishment planning.

1. Supplier capacity and reliability. A supplier who can produce 5,000 units a month cannot serve a demand plan of 8,000, whatever the sales team promises, and a supplier whose deliveries arrive late converts a correct plan into a stockout anyway.

2. Lead times. Order lead times determine how far ahead every replenishment decision must be made; production and transit lead times determine how early seasonal builds must begin.

3. Order constraints. Minimum order quantities, package sizes, and agreed order cycles shape the replenishment pattern regardless of how demand actually flows.

4. Working capital. Inventory is cash in another form. A pre-season build must be funded weeks before it sells, and the finance function's ceiling on stock investment is as hard a constraint as any warehouse wall.

5. Storage and handling capacity. Goods received must fit somewhere: space, racking, cold-chain capacity, and handling equipment all cap how much build-up is physically possible.

6. Transport capacity. Vehicle availability, sailing frequencies, and carrier allocations determine whether the planned volumes can actually move on schedule.

The supply plan that emerges is the demand plan translated into purchase orders, production schedules, warehouse space, and transport bookings that these constraints can genuinely deliver, or an explicit statement of what must change, in capacity, in timing, or in the demand plan itself, for the numbers to close. Which is exactly the work of the next block.

KEY CONCEPT

Sales and operations planning: one plan, agreed monthly

When the demand plan and the supply plan collide, someone must reconcile them, and an organisation that leaves this to hallway negotiations gets plans that optimise each function and fail the business. Sales and operations planning, usually abbreviated to S&OP, is a coordinated business planning process that aligns demand, supply, and financial objectives. It is a discipline formalised in the 1980s, and its core rules have not changed: the process runs typically as a monthly cycle, it plans at an aggregate, product-family level rather than item by item, it looks ahead up to about eighteen months, and it ends with a single plan that sales, operations, inventory, and finance all work from, one set of assumptions shared by every function.

The cycle is a sequence of reviews that builds one number at a time.

1. Data review. Historical demand, actual sales, inventory positions, and market intelligence are collected and cleaned, so the meeting argues about the future, not the arithmetic.

2. Demand review. The demand planning team proposes the demand plan, and sales, marketing, and market specialists test it against what the market is telling them; the output is one agreed demand number by family and month.

3. Supply review. Operations and procurement test that number against supply capacity, lead times, stock levels, and cost, and produce the supply plan that could serve it, or the gaps that prevent serving it.

4. Reconciliation. Where demand and supply disagree, options are assembled and costed: build earlier, shift promotions, add capacity, adjust the plan. This pre-meeting work clears the ground so that executives decide rather than negotiate.

5. Executive decision. Senior leadership approves the single plan, with its financial view of revenue, cost, and working capital, and issues it as the assumption base for the whole business.

S&OP is this lesson's answer to a question the methods alone cannot settle: a forecast model produces numbers, but only a conversation between the functions, and with key suppliers and customers, produces a plan the organisation can actually execute and afford.

KEY CONCEPT

Push, pull, and the time-phased plans

Position the forecast correctly and it does enormous work; misposition it and it manufactures the very errors it was meant to prevent. The vocabulary of positioning is push and pull. Push means executing against forecasts: goods are produced, purchased, or moved to where the forecast expects them to be needed. Pull means executing against actual demand: nothing moves until a real order signals it. Push offers availability and scale economies but carries the risk of every forecast error; pull carries almost no forecast risk but demands fast, reliable replenishment and accepts that some demand may wait.

Real supply chains are almost always hybrids. Somewhere in the chain sits a decoupling point: upstream of it the chain is pushed by forecast, producing and holding generic stock at aggregate level where error is relatively small; downstream of it the chain is pulled by actual orders, configuring and delivering what specific customers asked for. Choosing where that point sits, what is made to forecast and what to order, is among the most consequential planning decisions a business makes; postponement, later in this lesson, is the classic tool for moving it.

Above these execution decisions sit the time-phased planning concepts.

1. Material requirements planning, MRP. From a master schedule of what is to be produced, the bills of material listing what each product contains, and inventory records of what is already held, MRP computes time-phased orders for the dependent components: what to make or buy, in what quantity, and in which week, so that materials arrive when production needs them rather than all at once or too late.

2. Distribution requirements planning, DRP. DRP applies the same time-phased logic to distribution: from the demand plans of each distribution centre and the lead times between them, it computes when to ship what from where, replenishing the network in step with forecast demand instead of by ad hoc transfer.

Both are planning concepts, and both are only as good as the demand plan above them: MRP schedules the supply of components, DRP the deployment of stock through the network.

KEY CONCEPT

The bullwhip effect: small wobbles, large waves

Leave a supply chain to its own devices and a curious pathology appears: customer demand may be almost steady, yet orders to the factory swing wildly. This is the bullwhip effect, in which order variability amplifies upstream, so that each tier of the chain orders in swings larger than the swings it observed. A snap of the wrist at one end becomes a crack at the other. Researchers Lee, Padmanabhan, and Whang established the standard explanation in 1997: four causes, all perfectly rational at the level where they occur, which is why the effect is so hard to stop.

1. Demand-signal distortion. Each tier re-forecasts from the orders it receives, but those orders are already distorted by the tier below's batching and buffering, so each forecast overreacts and adjusts again, compounding the original wobble.

2. Order batching. Ordering weekly in small quantities is expensive, so buyers order fortnightly or monthly in large lots. The total is the same, but upstream sees lumpiness where demand was smooth.

3. Price fluctuations. Promotions and forward-buying pull orders forward: buyers stock up at low prices and starve the chain afterwards, so upstream mistakes the discount-driven spike for a change in real demand.

4. Rationing and shortage gaming. When supply runs short and sellers ration proportionally to orders, buyers inflate orders to secure share, cancelling once the scare passes; upstream sees phantom demand and builds capacity for it.

The costs land everywhere upstream: idle capacity one month, overtime and expediting the next, all without any change in what consumers actually wanted. The countermeasures attack the causes directly, as the worked example traces. Information sharing, passing point-of-sale sales data upstream so every tier forecasts from real demand rather than distorted orders, is the deepest remedy, and it is Lesson 1.2's definition of supply chain management, coordination and collaboration across companies, turned into daily practice. Smaller and more frequent orders flatten the batching; everyday stable pricing removes the forward-buying spikes; and vendor-managed inventory, in which the supplier replenishes directly from the customer's consumption data, removes a whole tier of distorted signals.

KEY CONCEPT

Postponement: shaping risk by delaying configuration

Forecasting reduces uncertainty; postponement reduces the cost of the uncertainty that forecasting cannot remove. Postponement, also called delayed differentiation, means keeping the product generic and deferring final configuration, dyeing, labelling, local packaging, final

assembly, until real demand is known. The classic teaching cases are knitwear dyed in the colours actually ordered after the season's shape of demand became visible, and paint that is mixed to the customer's chosen colour in the shop from a single generic base. No forecaster can predict how many litres of a specific shade a city will buy next month; every forecaster can estimate how many litres of base it will consume.

The mechanism is risk-shaping through aggregation. While the product is generic, one pool of stock serves many possible demands: the fifty colours a range might need require only the inventory of one base product plus local mixing, and the errors of individual colours partly cancel rather than accumulate. Forecast error is still present, but it is pushed up to the aggregated level where it is smallest, and the expensive, error-prone commitment, which colour, which label, which market, is made only when the demand information is freshest. This is why postponement suits products facing high demand uncertainty with a late, cheap configuration step: it pairs naturally with the push-pull positioning earlier in this lesson, moving the decoupling point downstream to just before differentiation.

Postponement is not free. It requires a configuration capability close to the customer, and holding products generic must not cost more than the mis-forecasts it avoids. The planner's judgement is a trade-off: the later the commitment, the smaller the risk and the later, and cheaper, the correction.

WORKED EXAMPLE

Six months of demand at a Kumasi distributor: moving average, exponential smoothing, and what the error means

A distributor in Kumasi supplies one fast-moving SKU of packaged foods to retail shops. Six months of recorded demand, in cartons, are 200, 210, 220, 230, 240, and 250 for months 1 to 6. All figures in this example are illustrative for teaching purposes. The planner must forecast month 7 and understand how wrong either method could be.

Step 1: The moving average. The month 4 forecast is the average of months 1 to 3: 630 divided by 3, or 210. Month 5 then forecasts at 220, and month 6 at 230, each the average of the three months before it. Actual demand turned out to be 230, 240, and 250: each moving-average forecast was exactly 20 cartons low.

Step 2: Exponential smoothing. With a smoothing weight of 0.3, and the month 1 forecast set at the first actual, 200, each new forecast is the previous forecast plus 0.3 of that period's error. The forecasts for months 2 to 6 work out to 200, 203, 208.1, 214.7, and 222.3, rounded to one decimal. Against the same actuals of 230, 240, and 250, these are low by 21.9, 25.3, and 27.7 cartons.

Step 3: Compare accuracy on the same window. To compare fairly, score both methods over the same three months, months 4 to 6. For the moving average, the absolute percentage errors are 8.7, 8.3, and 8.0 per cent (20 cartons against 230, 240, and 250), a MAPE of about 8.3 per cent.

For exponential smoothing the same computation gives about 10.4 per cent. The moving average fits this series better: a low smoothing weight trails a rising series further behind than a three-month average does.

Step 4: Read the pattern, not just the size. Every error, under both methods, had the same sign: too low, six times out of six. That is bias, not noise. Demand rose by 10 cartons a month, 25 per cent across the record, and both averaging methods lag a trend by construction. The remedy is not a larger safety stock but a better forecast: a trend-aware method such as regression on time, or the market intelligence that a new customer has just come on board.

Step 5: Translate error into stock policy. A persistent 20-carton under-forecast on a 250-carton month means a stockout or a premium emergency top-up at every replenishment cycle. Safety stock sized on the assumption of unbiased errors will keep missing in the same direction.

What does this example demonstrate? Error measurement is a diagnostic instrument: the MAPE sizes the miss, and the sign pattern names its cause.

Key lesson: fix bias in the forecast, buffer only noise with stock, and never compare methods on different windows.

WORKED EXAMPLE

One S&OP month at a consumer-goods importer in Tema

An importer in Tema brings in a family of packaged fruit-juice products and runs its sales and operations planning as a monthly cycle. The December meeting, held in October, walks the full loop. All figures in this example are illustrative for teaching purposes.

Step 1: Data review. The demand planning team assembles cleaned historical demand by product family, current stock positions, open orders, and market intelligence. The history shows a recurring festive-season lift: December sales have repeatedly run well above ordinary months, a seasonality the family plan must carry deliberately.

Step 2: Demand review. Sales and the demand planners agree the family plan: an ordinary month at 4,000 cartons, rising to a December demand plan of 6,500 cartons. One number, by family, is signed off and passed to supply.

Step 3: Supply review. Procurement and operations test the plan against constraints. The overseas supplier can ship at most 5,000 cartons a month, so even a full December shipment leaves 1,500 cartons to be supplied earlier and held. Lead time is eight weeks, which is why this meeting is happening in October. Storage space can absorb the build, but finance sets a working-capital ceiling that funds at most 1,200 cartons of extra stock: at an illustrative landed cost of GHS 40 per carton, about GHS 48,000 of additional cash tied up before the goods sell.

Step 4: Reconciliation. The gap is explicit: the December inflow of 4,000 cartons, the shipment already booked at the ordinary-month quantity and no longer enlargeable inside the eight-week

lead time, plus a 1,200-carton build covers 5,200 of the 6,500-carton plan, and opening stock adds about 300, leaving roughly 1,000 cartons short. The pre-meeting assembles costed options: a premium top-up shipment for part of the gap, moving part of the festive promotion into late November to pull demand forward, and concentrating the build on the family's fastest-moving SKUs so the shortage falls on the least damaging items.

Step 5: Executive decision. Leadership approves the 1,200-carton build within the working-capital ceiling, directs sales to shift a defined part of the promotion to late November, and schedules the decision on a premium top-up for the November cycle, when the December demand number will be re-checked against early-season sales.

What does this example demonstrate? S&OP is the mechanism that converts a forecast into a plan an organisation can actually fund, store, and ship, by forcing the demand number, the supply constraints, and the money into one room with the authority to trade them off.

Key lesson: the output of the cycle is not a better forecast but a single reconciled plan, one demand number, one supply answer, one financial view.

WORKED EXAMPLE

A bullwhip traced and tamed: from Hamburg's shops to a Shenzhen factory

One household product travels a chain: shops in Hamburg sell to consumers, a wholesaler in Rotterdam supplies the shops, and an importer orders from a factory in Shenzhen. The following figures are illustrative, chosen to show the mechanism plainly rather than as universal magnitudes.

Step 1: Measure the wobble at the customer. The shops' sales to consumers are steady: about 1,000 units a month, moving between 950 and 1,050. Consumers, buying one item at a time, barely change their behaviour at all.

Step 2: Watch batching distort the signal. The shops order from the wholesaler monthly, in whole boxes of 50, and round up for safety. Shop orders now run between 1,000 and 1,150. Nothing about consumer demand changed, but the order signal has widened.

Step 3: Watch forecast updating amplify it. The wholesaler re-forecasts from the shop orders it receives, not from consumer sales it cannot see, and after one late delivery it adds protective cover. Its orders to the importer now swing between 1,100 and 1,450.

Step 4: Watch shortage gaming finish the job. The importer, having once run out and lost sales, pads its orders and batches them: its orders to the Shenzhen factory swing between 900 and 1,900. A 100-unit wobble in shopping has become a 1,000-unit swing in factory orders, ten times the variation, with the four causes all visible: batching at the shops, demand-signal distortion at the wholesaler, shortage gaming at the importer, and a promotion calendar pulling orders forward on top.

Step 5: Apply the countermeasures. The three companies agree on two disciplines. Point-of-sale data from the shops is shared up the chain, so every tier forecasts from consumer demand rather than from distorted orders. Shops and the wholesaler move to smaller, more frequent orders in step with real offtake, and everyday pricing replaces deep promotions. Six months later, factory orders track consumption: the swings have narrowed toward the modest wobble the shopper always had.

What does this example demonstrate? Bullwhip is manufactured inside the chain by rational local behaviour, and it is unmade the same way: better information, smaller batches, and steadier prices.

Key lesson: each tier's defence against uncertainty becomes the next tier's problem; share the demand signal and the problem shrinks for everyone.

IMPORTANT NOTE

A forecast is an input to a decision, never a promise

The most damaging habit in planning is treating a forecast as a commitment that reality is expected to honour. A forecast is an input to decisions, produced to help someone choose a stock level, a booking, a build plan; actual demand is discovered only when customers act. Forecast demand and actual demand are different quantities, and the professional planner keeps them visibly separate in every review.

Because error is certain, mature organisations plan ranges rather than points. A demand plan carried as a single number invites a single, brittle supply plan; the same plan carried with its expected error band invites questions that matter: can we serve the upper end, and what happens to cost and stock at the lower end? The range is not vagueness. It is honesty about the error measured in this lesson, and it is what allows postponement, flexible capacity, and staged commitments to do their work.

The last discipline is the loop: every period, actuals are compared with forecasts, error is measured, and bias is hunted. A forecast that is never scored is not a plan but an opinion with seniority. And when the gap opens, remember where the cost lands: it never stays in the planning spreadsheet. It becomes a stockout for sales, an emergency shipment for transport, write-off for finance, and a queue for the warehouse, costs in exactly the families Lesson 10.1 catalogued. Forecast error is never free; somebody in the chain always pays it.

IMPORTANT NOTE

Improving supply planning accuracy: the conversation and the habits

When data is thin, a well-run sales and operations planning conversation beats the cleverest forecasting model. A model can only extend the patterns it is fed; the S&OP cycle adds what no history contains, what sales knows is closing, what the supplier can genuinely promise, what finance can fund, and forces those views into one reconciled number. The best organisations

use both: models for the arithmetic, the conversation for the judgement, and judgement always visible as an adjustment that can itself be scored.

Accuracy is built by habits more than by mathematics, and the basic measures are within reach of any serious operation.

1. Measure error every period. Track MAPE or MAD, and watch the sign of errors: bias is the cheapest error to fix and the most expensive to ignore.

2. Keep clean history. Record true demand, adding back lost sales where a stockout silenced it, so the data describes the market rather than the supply failures.

3. Forecast at two levels. Family and medium term for stability, item and short term for operations, with the hierarchy connected.

4. Share information along the chain. Point-of-sale data upstream is the deepest bullwhip remedy.

5. Order smaller and more often, and price steadily. Both flatten the signal every upstream tier must read.

6. De-risk the structure. Postponement and staged commitments shrink the cost of the error that remains.

None of these removes forecast error, which is why the earlier notes on buffers and ranges still stand. Together they shrink the error, expose the bias, and ensure that what error remains is carried deliberately, at the cheapest point in the chain.

Practice questions (Lesson Checks)

Practice Question 1

A distribution business in Accra is preparing to launch a new range of packaged shea-butter cosmetics. No sales history exists for the products, the range has never been sold in the market, and purchasing and warehouse space must be committed before launch. A colleague proposes applying a three-month moving average to last year's figures for a different product range.

Which option best describes the most appropriate forecasting approach?

1. Use the moving average on the different range's figures, because any numerical method is more reliable than judgement when launch decisions are pending.

2. Use qualitative forecasting: expert judgement from the sales team, structured market surveys of retailers and customers, and a Delphi-style expert panel, since no sales history exists for these products. ✓

3. Fit a regression linking demand to driver variables, using the new products' own sales data accumulated after launch, and delay all purchasing decisions until the model is available.

4. Forecast nothing before launch, because demand for a genuinely new product cannot be estimated and every commitment should simply be avoided.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

Qualitative forecasting, forecasting based on expert judgement, market knowledge, and other non-numerical information, is the method of necessity when history is absent. A new range with no sales record cannot be served by any technique that learns from past data, so the available evidence, expert knowledge, structured market surveys, and Delphi-style panels that aggregate independent expert opinions in anonymous, iterated rounds, is the honest basis for the launch commitment.

Answer 1 fails on the lesson's central rule: a moving average requires historical observations of the item being forecast, and last year's figures for a different range measure a different demand pattern. Answer 3 describes a sound method applied at the wrong time; regression needs data the new products do not yet have, and the business cannot postpone purchasing until after launch, because lead times force commitments now. Answer 4 abandons forecasting altogether, but refusing to forecast does not remove the decision; purchasing and space commitments will be made either way, merely without a stated demand assumption.

The takeaway: match the method to the data, and when there is no data, structure judgement instead of pretending history exists.

Practice Question 2

At a consumer-goods importer, the monthly demand review produces a festive-season demand plan of 6,500 cartons. The supply review then finds that supplier capacity, lead times, and a working-capital ceiling can realistically deliver about 5,200 cartons in December. Each function's plan is internally sound, but the two numbers disagree.

Which statement best describes what the sales and operations planning process should produce?

1. The demand plan stands unchanged, because the sales function owns the customer-facing number and operations must simply commit to matching it.

2. The supply plan stands unchanged, because physical capacity is the binding reality and sales must be told to reduce the number to whatever supply can deliver.

3. A single reconciled plan, approved at the executive meeting, that balances the demand number against supply capacity, inventory, and the financial view, with the trade-offs, a pre-build, a shifted promotion, an adjusted plan, chosen explicitly. ✓

4. Both plans are kept separately and each function works to its own number, with the differences noted and reconciled when the year-end accounts are prepared.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

Sales and operations planning exists precisely for this collision. Its cycle, data review, demand review, supply review, reconciliation, and executive decision, forces the demand number, the supply constraints, and the financial view into one process that ends with a single plan: one demand number, one supply answer, one financial view, shared as the assumption base by every function. In the lesson's worked example, the reconciliations were concrete and costed, a funded pre-build, a promotion shifted to pull demand forward, a decision deferred until the next cycle could re-check the number, and the executive meeting chose among them.

Answer 1 lets the commercial plan ignore hard constraints, guaranteeing stockouts and emergency spending downstream. Answer 2 lets operations veto demand, quietly converting a planning disagreement into a revenue decision taken without the commercial function. Answer 3's alternative, Answer 4, preserves the silo the process exists to dismantle: two internally sound plans that disagree produce purchases against one number and sales against another, and the difference surfaces later as cost.

The takeaway: S&OP's product is not a winning plan but a single reconciled one, agreed at the level with the authority to trade demand, capacity, and money against each other.

Practice Question 3

A factory manager notices that orders arriving from an importer swing far more than the sales reported by the shops at the end of the chain. After the companies begin sharing point-of-sale data, ordering in smaller and more frequent quantities, and replacing deep promotions with steady everyday pricing, the swings in factory orders narrow substantially.

Which statement best explains both the original pattern and the improvement?

1. Consumer demand is inherently more volatile than retail sales data suggests, so the factory swings were real and the improvement came from demand genuinely steadying.

2. Transport capacity along the chain varied between months, causing irregular deliveries and therefore irregular ordering, and the improvement came from better vehicles.

3. Order variability amplifies upstream through demand-signal distortion, order batching, price fluctuations, and shortage gaming, and the remedies worked because they attack those causes: shared sales data, smaller and more frequent orders, and stable pricing. ✓

4. Warehouse space at the wholesaler was too small, forcing irregular ordering, and the improvement came from building additional storage capacity.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

The bullwhip effect is the amplification of order variability upstream: each tier orders in swings larger than the swings it observed, even when consumer demand barely moves. The lesson named the four standard causes established by Lee, Padmanabhan, and Whang: demand-signal distortion, where each tier re-forecasts from already-distorted orders; order batching, which turns smooth sales into lumpy orders; price fluctuations, which pull orders forward in promotions; and rationing or shortage gaming, where buyers inflate orders to protect supply. The countermeasures in the scenario attack the causes directly, sharing point-of-sale data so every tier forecasts from real demand, smaller and more frequent orders to flatten batching, and everyday pricing to remove forward-buying spikes.

Answer 1 mistakes the amplified orders for real demand, the exact misreading that leads factories to build capacity for phantom swings. Answers 2 and 4 attribute the pattern to physical constraints, vehicles and storage, but the scenario's remedy contained no new physical capacity at all: information and ordering discipline alone narrowed the swings, which only the demand-signal explanation accounts for.

The takeaway: when upstream orders swing harder than downstream sales, look inside the chain for the four causes, and remedy them with shared information and smoother ordering.

Fact-check appendix (audit notes — not for entry)

Definitions of Demand Forecasting, Supply Planning, Historical Demand, Qualitative Forecasting, Quantitative Forecasting, Moving Average, Seasonality, Forecast Accuracy, Forecast Error, Demand Variability, Lead Time, Supply Capacity, Sales and Operations Planning (S&OP), and Demand Planning follow the lesson brief's key terms, defined at first use and used naturally in the blocks. — source: curriculum/lesson_briefs/module-13/lesson-13.1.md key terms.

Forecasting families taught as: qualitative (market survey as structured customer/channel expectations; Delphi as iterated anonymous expert panels) and quantitative time series (moving average as mean of last n periods; exponential smoothing as weighting recent data more; regression/causal as demand explained by driver variables). — source: research sheet logistics-scm-frameworks.md section 8.

MAPE defined as the average of absolute error divided by actual demand, expressed as a percentage; MAD as the average absolute error in demand units; MAPE described as fragile near zero actuals. No accuracy benchmarks (no "good" MAPE thresholds) are stated anywhere. — source: research sheet logistics-scm-frameworks.md section 7 and DO NOT CLAIM list.

S&OP taught as a cross-functional, typically monthly process combining demand forecasting with supply planning so sales, operations, inventory, and finance work from one set of assumptions, at product-family/aggregate level, with a horizon up to about 18 months; the cycle presented as data review, demand review/plan, supply review/plan, reconciliation (pre-meeting), executive meeting; described as formalised in the 1980s without naming individuals or bodies. — source: research sheet logistics-scm-frameworks.md section 8.

Push (execute against forecasts) versus pull (execute against actual demand), with real chains usually hybrid around a decoupling stock point. — source: research sheet logistics-scm-frameworks.md section 8.

MRP as time-phased orders for dependent-demand components from master schedule, bills of material, and inventory records; DRP as the same time-phased logic applied to replenishing distribution centres; both taught as planning concepts, with no product or vendor names. — source: research sheet logistics-scm-frameworks.md section 8.

Bullwhip effect: order variability amplifies upstream; four causes attributed to Lee, Padmanabhan, and Whang (1997): demand signal processing (forecast updating), order batching, price fluctuations, rationing/shortage gaming. Countermeasures taught as information sharing (point-of-sale data), smaller/more frequent orders, everyday stable pricing, and vendor-managed inventory, all per the sheet. No amplification percentages are stated as fact; all figures in the bullwhip worked example are expressly illustrative of the mechanism. — source: research sheet logistics-scm-frameworks.md section 8 and DO NOT CLAIM item 16.

Postponement (delayed differentiation): keep the product generic and defer final configuration (dyeing, labelling, local packaging, final assembly) until real demand is known; suited to high demand uncertainty; classic cases referenced only as the sheet gives them (late-dyed knitwear; in-store paint mixing), with no company case-study claims beyond the sheet. — source: research sheet logistics-scm-frameworks.md section 8.

Reorder point taught as expected demand during the lead time plus safety stock, consistent with Lesson 8.3's inventory control content; safety stock as buffer against demand and lead-time variability; service level as target probability of no stockout, with higher targets needing disproportionately more safety stock (no formulas presented as universal). — source: research sheet logistics-scm-frameworks.md section 5; curriculum/lesson_briefs/module-08/lesson-8.3.md.

Lead time taught as order-to-receipt including order processing, production/pick-pack, handling, transport, and customs clearance internationally. — source: research sheet logistics-scm-frameworks.md section 7.

Forecasting software and machine-learning forecasting mentioned only as a qualitative trend automating the same principles, with no performance, ROI, or accuracy claims and no product names. — source: research sheet logistics-scm-frameworks.md section 9 (AI/ML uses, qualitative).

All arithmetic (moving average, exponential smoothing with weight 0.3, MAPE of about 8.3 per cent versus about 10.4 per cent over the common months 4-6 window; S&OP carton build and GHS figures; bullwhip order ranges) is computed from expressly illustrative teaching figures labelled as such in each worked example; no real-company data is used. — source: WRITING_SPEC.md fact-safety rule 5.1.

Companies in examples are generic ("a distributor in Kumasi", "an importer in Tema", shops in Hamburg, a wholesaler in Rotterdam, a factory in Shenzhen); no factual claims are attached to real named firms; Ghana and international (Hamburg-Rotterdam-Shenzhen) settings satisfy the example-placement rule. — source: WRITING_SPEC.md section 4.

Cross-references by name, no re-teaching: Lesson 1.2 Logistics versus Supply Chain Management (supply chain coordination across companies); Lesson 8.3 Inventory Control, FIFO, and FEFO (reorder point and safety stock); Lesson 10.1 Components of Total Logistics Cost (cost families absorbing forecast error); forward references Lesson 13.2 Transport Route and Network Optimisation, Lesson 13.3 Inventory Optimisation and Service Levels (service levels and

safety-stock optimisation), Lesson 13.4 Logistics Performance Indicators and KPIs. — source: WRITING_SPEC.md section 7; lesson briefs for modules 1, 8, 10, and 13.

Lesson 13.2 — Transport Route and Network Optimisation

Summary

This lesson examines how transportation routes and logistics networks are designed and improved to achieve efficient cargo movement. Learners will explore route selection, distribution networks, vehicle capacity, delivery scheduling, transportation costs, and service requirements. The lesson also introduces network optimisation methods and explains how organisations balance cost, transit time, reliability, and operational constraints when planning transportation activities.

Learning objectives

- Define transportation route optimisation and logistics network optimisation.
- Explain the importance of efficient route planning in logistics operations.
- Identify factors influencing transportation route selection.
- Describe the components of a logistics distribution network.
- Distinguish between direct delivery and hub-and-spoke distribution arrangements.
- Explain how vehicle capacity and delivery schedules affect route planning.
- Identify common transportation constraints and operational trade-offs.
- Describe how route optimisation can reduce transportation costs and delivery delays.
- Explain the role of digital mapping and optimisation software in transportation planning.
- Evaluate basic transportation alternatives using cost, distance, time, and service requirements.

Key terms

Route Optimisation — The process of identifying transportation routes that best satisfy specified cost, time, distance, and operational requirements.

Logistics Network — A system of facilities, transportation links, and operational relationships used to move and distribute goods.

Distribution Network — The arrangement of warehouses, distribution centres, transport routes, and delivery points used to supply customers.

Direct Delivery — A distribution arrangement in which goods move directly from an origin to a destination without passing through an intermediate distribution facility.

Hub-and-Spoke Network — A distribution structure in which shipments move through central hubs connecting multiple origins and destinations.

Vehicle Routing — The process of assigning delivery or collection locations to vehicles and determining their travel sequences.

Vehicle Capacity — The maximum permitted or practical quantity, weight, or volume of cargo that a vehicle can transport.

Load Factor — A measure of the proportion of available transport capacity being utilised.

Delivery Window — A specified period during which goods are expected or permitted to be delivered.

Route Constraint — A restriction affecting route selection, such as vehicle capacity, road access, time limits, or delivery requirements.

Network Optimisation — The improvement of facility locations, transportation links, and operational flows to achieve defined performance objectives.

Transportation Planning — The process of organising transport resources, routes, schedules, and services to meet shipment requirements.

Backhaul — A return or subsequent transport movement that may be used to carry additional cargo rather than operating empty.

Last-Mile Delivery — The final transportation stage in which goods reach their intended recipient.

Teaching content

KEY CONCEPT

Route decisions and network decisions are two different questions

This lesson opens the optimisation layer of transportation planning. Module 3 taught how transport modes are chosen, and Lesson 3.2 defined the multimodal and intermodal arrangements this lesson revisits at planning level. Lesson 7.1 and 7.4 covered ocean operations and freight charges; Lesson 12.1, International Transport Corridors, followed goods along border-crossing routes. This lesson stands above those operational views and asks the planning questions: which way should this shipment travel, and what structure should the company build so that the shipments of the next five years travel well? Lesson 13.1, Demand Forecasting and Supply Planning, supplied the demand picture both answers depend on; the next lesson, 13.3, Inventory Optimisation and Service Levels, will optimise the stock the network holds.

Transportation planning is the process of organising transport resources, routes, schedules, and services to meet shipment requirements. Within it sit two decisions that differ in kind, and confusing them is the first mistake this lesson corrects.

1. Route optimisation is the process of identifying transportation routes that best satisfy specified cost, time, distance, and operational requirements. It answers the question: given the

network as it exists, which way should this consignment go? Route decisions are made daily, at the dispatch desk, shipment by shipment.

2. Network optimisation is the improvement of facility locations, transportation links, and operational flows to achieve defined performance objectives. It answers a different question: what should the network itself look like? A logistics network is the system of facilities, transportation links, and operational relationships used to move and distribute goods — warehouses, hubs, ports, routes, and the services connecting them. Network decisions are investments: they cost capital, take months to implement, and are revisited rarely.

The clocks differ. A route can be changed next week at little cost beyond a change of mind; a distribution centre cannot. That difference in reversibility is why the two decisions need different data, different approvals, and different review cycles — and why this lesson keeps them separate throughout.

KEY CONCEPT

What a route comparison actually weighs

Route selection at planning level is a comparison of complete door-to-door options, not a scan of leg-by-leg prices. Lesson 7.4 taught the discipline of reading the whole charge build, and Lesson 10.1, Components of Total Logistics Cost, the total-cost lens; applied to a route, the lens asks what each option truly costs once everything it causes is counted. The comparison weighs six inputs.

1. Cost per leg. Each leg's linehaul, fuel, handling, and accessorial charges, built into a door-to-door figure — including the empty positioning legs vehicles run to reach their loading points.

2. Transit time. Door to door, including expected waiting at ports, terminals, and borders — not driving time alone; Lesson 12.1 showed how waiting clusters at a corridor's seams.

3. Reliability. How tightly transit times cluster around their average. Border friction, met in Lesson 12.1, turns a short routing into an unreliable one, and unreliability forces buffers of time and stock downstream.

4. Handling count. Every touch — a reload at a terminal, a transfer between vehicles, a cross-dock — adds delay risk and damage risk. This is Lesson 3.3's handling principle restated at planning level: the more a consignment is touched, the more can go wrong to it.

5. Constraints. A route constraint is a restriction affecting route selection, such as vehicle capacity, road access, time limits, or delivery requirements. Terminal cut-offs, vehicle size limits on certain roads, and equipment availability at the right place and date can disqualify an option before price is discussed.

6. The delivery window. A delivery window is a specified period during which goods are expected or permitted to be delivered. A routing arriving outside the window may be as useless as one arriving a day late; the buyer's receiving rhythm, not the map, sets the real deadline.

Route comparison also reopens the mode question of Lesson 3.2: a routing may differ from its rival by mode, not merely by road. Combining modes under one responsible operator is the multimodal arrangement; moving goods in one unchanged loading unit, such as a container, across modes is the intermodal arrangement. Both matter practically: each transfer between modes is a touch, and the contract structure decides who answers for the journey. A routing is finally chosen by weighing all six inputs together — never by picking the cheapest single leg.

KEY CONCEPT

The components of a distribution network

A distribution network is the arrangement of warehouses, distribution centres, transport routes, and delivery points used to supply customers. It is the network seen from the customer's end: the fixed structure through which every order must pass. Its components recur in every industry.

1. Origins. The factories, supplier sites, and import ports where stock enters the network. For an importing business the origin may be a gateway port such as Tema; for a manufacturer, the factory gate.

2. Storage and flow facilities. Central warehouses holding national stock, regional distribution centres holding stock close to demand, and cross-dock terminals, where goods move from the inbound vehicle to the outbound vehicle with little or no storage in between. The consolidation machinery of Lesson 5.3 and the inland depots of Lesson 9.2 live at this layer. Each facility is a place where stock rests and where handoffs happen.

3. Transportation links. The scheduled or ad hoc services joining the facilities: linehauls between warehouse and distribution centre, collection routes from suppliers, delivery rounds to customers.

4. Delivery points. The shops, sites, and households where goods leave the network. The final stage has its own name: last-mile delivery is the final transportation stage in which goods reach their intended recipient. Loads are small and stops are many, so each journey's cost is spread over little cargo and cost per unit delivered runs high — which is why the location of stock relative to customers matters.

Two facts make the network the master decision of this lesson. First, stock lives in the network: where facilities sit decides how much inventory the company must hold and where. Second, handoffs happen in the network: every facility a consignment passes through is a touch, carrying the delay and damage risk established earlier. Change the network and you change the cost and service of every shipment that will ever run through it — which is why network decisions are taken deliberately and rarely, and route decisions daily within them.

KEY CONCEPT

Direct delivery and hub-and-spoke distribution

Two pure structures anchor every distribution network discussion.

Direct delivery is a distribution arrangement in which goods move directly from an origin to a destination without passing through an intermediate distribution facility. A hub-and-spoke network is a distribution structure in which shipments move through central hubs connecting multiple origins and destinations. Each structure wins under different flow conditions; knowing which hold is the whole skill.

1. Direct delivery wins where flows are large and regular. When one origin-destination pair generates enough volume to fill vehicles on its own, a direct line gives the shortest distance, fewest touches, fastest transit, and no hub cost or delay. A full-truckload shuttle between a factory and a major customer is the classic case.

2. Hub-and-spoke wins where many origins meet many destinations in small quantities. When no single pair generates a full load, direct lines run half empty or needlessly often; a hub collects many small consignments, sorts them, and dispatches each outbound vehicle full. This is the consolidation logic of Lesson 5.3 expressed as a network shape, and the standard design of part-load road networks, whose carriers combine several shippers' freight in one vehicle — cheaper per unit, more handling, longer transit — which is why they typically operate hub-and-spoke terminal networks.

The hub's price is paid in distance, touches, and time: every inter-town consignment travels via the hub, is handled there, and waits for the outbound departure. The direct line's price is paid in utilisation: vehicles run part full unless the flow justifies them.

Few real networks are pure. The working pattern is hybrid — direct lines where flows are dense, a hub-and-spoke layer where they are thin; worked example two puts numbers on the choice. The habit is to let the flow pattern, not fashion, decide which lanes hub and which run direct.

KEY CONCEPT

Consolidation, vehicle capacity, and load factor

Route and network choices are made real by vehicle choices, and vehicle economics rest on two measures.

Vehicle capacity is the maximum permitted or practical quantity, weight, or volume of cargo that a vehicle can transport. It is bounded from two sides: the practical side, how much the body holds, and the legal side, permitted weight limits enforced at weighbridges along major corridors. Load factor is a measure of the proportion of available transport capacity being utilised. A full vehicle runs at a high load factor; a vehicle carrying one pallet on a twelve-pallet body runs at a twelfth of its capacity.

Load factor is the quiet lever behind most freight economics and a major one behind transport emissions, because a moving vehicle burns fuel largely whether it is full or not; raising it spreads the same journey's cost, and the same emissions, over more cargo.

Consolidation is the deliberate raising of load factor, in recognisable forms.

1. Full-truckload movement dedicates a whole vehicle to one shipper's load: direct, fast, minimal handling, and the sensible choice where the flow fills the vehicle.

2. Shared part-load movement — less-than-truckload shipping, called groupage in freight forwarding — combines several consignments in one vehicle: cheaper per unit, more handling, longer transit, and normally organised through a hub network for the reasons just given.

3. Scheduled collection routes, of which the milk run in worked example three is the pattern: one vehicle on a fixed loop consolidating small loads from several points into one journey.

The link back to route planning is direct: vehicle capacity decides how many vehicles a day's demand requires and which routings are physically possible; load factor decides whether each chosen routing is economic. A comparison that ignores both compares fantasies — the cheapest quoted option is no bargain if it cannot legally or physically carry the load, and no routing is cheap running at a tenth of its capacity.

KEY CONCEPT

Vehicle routing and delivery schedules

Vehicle routing is the process of assigning delivery or collection locations to vehicles and determining their travel sequences. It is route optimisation at the finest grain: not which corridor or mode, but which vehicle visits which stops in which order. Efficient route planning matters because distance, fuel, driving hours, and missed windows are all decided at this grain: a poor sequence adds hours to a day's work without a single road closing, while a well-planned round cuts transport cost and delivery delays with the vehicles the company already owns — one of the few optimisations available every day at almost no capital cost.

Three realities shape every routing plan.

1. Vehicle capacity and stop count. A vehicle can serve only as many stops as its capacity and its working day allow. Overload the round and the plan fails at the third stop; underuse it and the round wastes capacity already paid for.

2. Delivery windows. Customer receiving hours, urban delivery restrictions at certain times, and port or terminal cut-offs carve the day into permitted slots. A sequence that ignores them produces arrivals nobody can receive — and a failed delivery is the most expensive stop of all, its cost paid twice, in the wasted journey and the re-run. A sequence built around windows looks longer on the map and finishes sooner in practice.

3. Sequencing under constraints. Stops are ordered to minimise distance and driving time while respecting windows, vehicle access limits, and load separation requirements. Driving and rest rules bound how far a vehicle and crew may legally travel in a period, which is why some journeys are planned as overnight stops rather than same-day runs.

Done well, routing is where optimisation visibly reduces transport costs and delivery delays: fewer kilometres, fuller vehicles, arrivals inside promised windows, fewer failed deliveries and re-runs. Done as an afterthought at the dispatch desk each morning, it quietly gives back everything the network design saved.

KEY CONCEPT

Backhaul and the cost of running empty

A backhaul is a return or subsequent transport movement that may be used to carry additional cargo rather than operating empty. Its opposite is empty running: a vehicle travelling without paying cargo. A truck sent from Accra to Kumasi that returns the same day carrying nothing costs its owner the return journey's fuel, driver time, and vehicle wear, and earns nothing against them; the price of the outbound load quietly carries the empty return inside it.

Empty running is a major cost and emissions inefficiency. The emissions point deserves to stand beside the money: an empty leg produces emissions with no freight to show for them, and reducing empty running counts among the recognised levers of greener transport, beside shifting to less emissive modes and improving load factors.

Why does empty running persist? Flows are unbalanced in space and time: goods move from production regions to consumption regions, and the return direction offers less cargo than the outbound. A network's geography therefore decides how hard backhauls are to find — making backhaul potential itself a network-design consideration.

Planning return loads is a transport-planning discipline, not a stroke of luck.

1. Match return cargo where the flows exist: a distribution run can backhaul supplier collections, empty pallets, or reusable packaging along the same corridor.

2. Trade capacity in both directions: carriers will generally price an outbound lane better where the shipper can offer them return traffic, relieving their own empty-running problem.

3. Accept the second-best backhaul where the perfect one does not exist; a part-load return still beats an empty one on both cost and emissions.

Worked example three shows the milk run as a structural answer to empty legs, and worked example one shows how positioning legs change a route comparison on a cross-border lane.

KEY CONCEPT

Designing the network: the distribution-centre question

The central network-design question is deceptively simple: how many distribution centres does the business need, and where should they sit? Beneath it lies the best-established trade-off in network design, and every network decision is a position on that trade-off.

More distribution centres mean each facility sits closer to its customers. Outbound delivery distances fall, delivery becomes faster and response to orders quicker, and the last mile shortens. But each additional facility brings fixed cost — premises, equipment, systems, staff — and, less obviously, it fragments the company's inventory: each site must hold its own buffer against its own local demand, so total stock rises for the same level of protection. A centralised network does the reverse: one pool of stock serves everyone at the lowest total inventory, but delivery distances, transit times, and last-mile costs grow.

The design questions that decide a sound position on the trade-off are the ones this module has assembled.

1. Which markets must be served, and where do they sit? Geography sets the delivery distances any structure must produce.

2. What is the service promise? A next-day promise forces stock close to customers; a slower promise lets a central pool serve widely.

3. What is the demand pattern? The forecast built in Lesson 13.1 tells the planner where volume concentrates and how it varies — the raw material of the fragmentation calculation in worked example four.

Network design also extends to how products travel through the structure. Postponement — keeping the product generic and deferring final configuration, such as labelling or final packing, until real demand is known — can matter as much as facility count where demand is uncertain and varied, because it lets one central stock serve many possible endings.

The headline: more distribution centres buy service with inventory and fixed cost; fewer buy inventory efficiency with distance and time. Neither pole is correct; the service promise and the demand pattern decide the price worth paying.

KEY CONCEPT

Digital mapping and optimisation software in transportation planning

None of this lesson's comparisons are new; software changes the scale at which they can be made and remade. Digital mapping supplies the geography — road networks, distances, travel times, access restrictions — that turns which way into a computable question. On that geography sit two families of tools.

1. Routing and scheduling tools work on the vehicle-routing problem family described earlier: given stops, quantities, windows, capacities, and driving-time limits, they propose vehicle assignments and stop sequences, and re-propose them when an order arrives or a window changes. Around those plans, a transport management system — the software class that plans and executes transport — typically adds carrier selection and rating, tendering, track-and-trace, and freight audit and settlement.

2. Network design tools take the longer view: given demand by region, facility costs, and transport rates, they compare candidate structures — two distribution centres against three, this region served from here or there — on total cost and service before capital is committed.

Two cautions belong in the same breath as the capability. First, these tools do not remove judgement; they rank options by the inputs given, and the next block takes up what happens when the inputs are wrong. Second, published claims about the savings such tools deliver are vendor-dependent and cannot be treated as facts; what can be taught is what the tools do — compute, compare, and re-compare at a scale and speed no manual planner can match, so that re-planning becomes cheap enough to do routinely rather than rarely. The professional standard is to know what question each tool answers and what data it needs. Such a tool earns its keep as a fast, consistent calculator of comparisons the planner defines — never as the planner's replacement.

KEY CONCEPT

The honesty of optimisation: models rank the inputs they are given

Optimisation, whether performed in software or on the back of two quotations, has one non-negotiable property: it ranks options by the inputs given. Change the inputs and the ranking changes. The computing phrase for the failure mode is garbage in, garbage out: a model fed optimistic transit times, incomplete handling costs, and last year's fuel prices confidently returns an answer that is precisely wrong.

Three disciplines keep optimisation honest.

1. Feed it the whole cost. Service and risk are part of cost. A cheap fragile route is not cheap: the routing that saves money on linehaul but adds two handling touches carries damage and delay risk that belongs inside the comparison, as worked example one demonstrates. Reliability belongs in the cost figure too: the buffers a company holds against an unpredictable lane are real money spent because of that lane.

2. Optimise the total system, not the silo. Lesson 10.1's total-cost lens, applied at network scale, is the discipline here: the cheapest transport option is frequently the most expensive supply chain decision, because slow and unreliable transport raises inventory, expediting, and stockout costs elsewhere in the operation. The question is never only whether a leg is cheap, but what the choice costs the whole system.

3. Respect the model's simplifications. Every network model compares a simplified world: average demands, average rates, the scenarios someone chose to run. The output is a structured comparison of assumptions, not a prophecy. Read it by asking which assumption, if wrong, would most change the answer.

Held together, the three disciplines give optimisation its proper role: a rigorous, repeatable way to compare options the planner has framed well, with the planner — not the algorithm — accountable for the framing. That is the advanced standard this lesson sets: optimise deliberately, feed the comparison honestly, and judge the outputs as a professional judges any calculation.

WORKED EXAMPLE

Choosing between two routings: a recurring Kumasi to Ouagadougou consignment

A Ghanaian distributor of beverages and household goods ships a recurring weekly consignment — 24 pallets — from its Kumasi warehouse to a buyer in Ouagadougou, Burkina Faso. The buyer receives goods only within an agreed weekly delivery window. Two routings are on the table. All figures in this example are illustrative for teaching purposes.

Step 1: Describe the options.
Option A runs north from Kumasi on Ghana's paved axis toward the northern border crossing with Burkina Faso, then on to Ouagadougou. It crosses one international border, moves sealed under the regional transit arrangement met in Lesson 12.1 — one document, no reloading en route — and carries an all-in door-to-door quote of GHS 40,000. Typical transit is three days, and the trip record shows the delivery window missed once in the last ten trips.
Option B runs south-east to the coast, crosses at the coastal border into Togo, and runs north on the paved axis linking Lomé with Ouagadougou. It crosses two international borders and adds one intermediate handling: at a Togo terminal the consignment is transferred between carriers. Its quoted price is GHS 37,500. Typical transit is four days when both borders move well, six when they do not, and the delivery window has been missed on about half the trips.

Step 2: Price the differences.
The second border and the coastal detour add, on average, two extra driver-and-vehicle days per trip, at an illustrative GHS 1,500 per day: GHS 3,000. The terminal transfer adds a handling touch, with its damage and delay risk. Missed windows trigger an average penalty of GHS 6,000 per miss in discounts, waiting, and storage. Option A expects 0.1 x 6,000 = GHS 600 per trip; Option B expects 0.5 x 6,000 = GHS 3,000.

Step 3: Compare door to door.
Option A: 40,000 + 600 = GHS 40,600, reliably. Option B: 37,500 + 3,000 + 3,000 = GHS 43,500, variably. The lower quote is GHS 2,900 per trip dearer in practice — and less predictable besides, forcing the buyer to hold buffer stock against the bad case.

What does this example demonstrate?
The linehaul quote was one line of a seven-line comparison. The borders, the extra touch, the reliability record, and the delivery window decided the result, as the route-comparison inputs predict.

Key lesson
Compare routings door to door, on total expected cost including waiting, handling, and missed windows, weighted by the buyer's tolerance for unreliability. Different numbers could reverse the result: cheaper coastal rates and a dependable second border could make Option B win. The method, not this answer, is the lesson.

WORKED EXAMPLE

Hub-and-spoke versus point-to-point: the arithmetic of five towns

A European distributor operates a national distribution centre at Rotterdam and serves five regional towns. Flows between the towns are scattered: each pair exchanges, on average, one-fifth of a truckload per day. Management asks whether the towns should be linked by direct lines or through a regional hub. All figures in this example are illustrative for teaching purposes.

Step 1: Count the direct option.
Connecting every pair of towns by a direct line needs one route per pair: five towns, each paired with four others, counted once, give 5 x 4 / 2 = 10 routes. Each route carries only its own pair's one-fifth of a load, so every direct vehicle runs at roughly a 20 percent load factor. Ten mostly empty vehicles, ten mostly empty return legs, with the cost and emissions of both.

Step 2: Count the hub option.
Insert a hub between the towns. Each town now has a single spoke to the hub. Each spoke carries the town's combined traffic with the other four towns: 4 x one-fifth = four-fifths of a load inbound toward the hub, and the same outbound from it. Connections fall from 10 routes to 5 spokes, and each spoke movement runs at roughly an 80 percent load factor instead of 20. The price: every inter-town consignment now travels two legs, is handled at the hub, and waits for the outbound departure, while the hub facility itself costs money to run.

Step 3: Find where each wins.
For the scattered flows the hub wins clearly — half the connections, four times the load factor, fuller vehicles, lower cost per unit carried. This is the arithmetic behind part-load networks, and why such carriers typically run hub-and-spoke terminal networks. But suppose two of the towns, an hour apart, begin exchanging two full truckloads every day. Hubbing that flow turns the two daily full loads into four spoke movements and adds two hub handlings per load to cargo that could travel direct, full, in one leg. The dense lane justifies its own direct line, and the hub keeps serving the thin flows around it.

What does this example demonstrate?
The arithmetic behind structure choice: count the flows. Many small flows between many points

favour a hub-and-spoke network; dense, regular flows between a pair favour direct delivery. Volume per route, not preference, decides.

Key lesson
Direct delivery and hub-and-spoke are answers to different flow patterns; durable networks are hybrid, each lane running on the structure its volume supports, reviewed as volumes change.

WORKED EXAMPLE

The milk-run: one loop replaces four collection trips

A fruit-juice packer in Tema receives packaging materials daily from four suppliers around the industrial area — cartons, bottles, labels, and film — each supplier dispatching four pallets a day to the factory. All figures in this example are illustrative for teaching purposes.

Step 1: Describe the old arrangement.
The factory booked four dedicated collections a day. Each collection used a 12-pallet truck to carry 4 pallets — a one-third load factor — and each truck returned to its base empty. Four trips, four collection charges, and two-thirds of the capacity paid for sitting idle. The average charge was GHS 1,800 per collection: 4 x 1,800 = GHS 7,200 a day. Arrival times varied with each carrier's day, so the warehouse could not plan receiving staff around them, and late collections pushed the packing schedule back.

Step 2: Design the loop.
The packer moved the collections onto a milk run: one vehicle on a fixed, repeated circular route with multiple pickup stops. A single 16-pallet vehicle now runs a published loop each morning, visiting the four suppliers in a fixed sequence, collecting 16 pallets, and returning full. Load factor: 16 / 16 = 100 percent. One trip replaces four, and four empty return legs collapse into one loaded return. The loop contract costs GHS 3,500 a day (illustrative), a saving of GHS 3,700 a day over the four dedicated collections — and because the loop runs to a fixed timetable, material arrivals become predictable, which the packing line had never had.

Step 3: Build in the constraints that make it work.
The loop functions only because three disciplines hold. First, each supplier prepares its four pallets inside a fixed pickup window, so the vehicle never waits and the sequence never slips. Second, the total load of 16 pallets sits within the vehicle's capacity, and the full loop fits inside the driver's permitted working day. Third, because one vehicle carries four supply lines, the packer holds a contingency — a backup vehicle agreement and a rule to resequence or split the loop when the truck fails — so one breakdown cannot stop four production inputs at once.

What does this example demonstrate?
The milk run consolidates small loads into one scheduled route: fewer trips, higher load factor, lower cost, and more predictable arrivals than the ad hoc collections it replaced — consolidation, capacity planning, and scheduling in a single design.

Key lesson

Fixed loops trade flexibility for cost and reliability. They work when pickup windows are honoured, the load fits the vehicle, and a contingency exists for the loop's single point of failure.

WORKED EXAMPLE

One warehouse or two: the distribution-centre question

A pharmaceutical distributor in Accra supplies pharmacies and hospitals across Ghana from one national warehouse. Northern customers around Tamale are replenished by long road journeys from Accra, with lead times of days, and the service promise in the north is slipping as sales grow. The board considers a second distribution centre in Kumasi. All figures in this example are illustrative for teaching purposes.

Step 1: Quantify the stock effect.

Today the single warehouse holds a national buffer of 6,000 units of the fast-moving lines (illustrative). Two sites do not hold 3,000 each for the same protection. Each site must carry its own buffer against its own region's uncertain demand, even when the other region is quiet. The planner estimates 4,200 units in Accra plus 3,000 in Kumasi — 7,200 where one pool held 6,000, roughly a fifth more — and the two sites also hold separate working stock, so total inventory rises further (illustrative). This is inventory fragmentation: splitting one pool into sites multiplies the buffers, as the network-design trade-off predicts.

Step 2: Quantify the service effect.

From Kumasi, replenishments to the north shorten by roughly a day's drive; Kumasi-area customers move to same-day collection and delivery; and Tamale's lead time falls materially. Outbound transport cost per unit falls with the shorter distance. The second site also adds resilience: a disruption at either site no longer strands the whole country's supply.

Step 3: Weigh the trade-off.

Against the service gains stand the second facility's fixed costs — premises, staff, systems, management attention — and the extra fifth of inventory with its carrying cost. The decision turns on this lesson's questions. What service promise does the business actually sell — can northern customers be served in full and on time from Accra at all? What does Lesson 13.1's demand forecast show — is northern volume large and growing enough to feed a distribution centre? And what do the transport rates between the sites allow? If northern demand were thin and slow-moving, the same money might serve better as a scheduled delivery service from the central warehouse.

What does this example demonstrate?

More distribution centres buy service, shorter last-mile distance, and resilience — paid for in fixed cost and fragmented, larger total stock. Fewer buy inventory efficiency — paid for in distance and time. The right count follows the service promise and the demand pattern, not the real estate available.

Key lesson

Network design is a cost-versus-service decision made with demand data. Every added facility must be paid for in stock and fixed cost, and justified in service; the demand forecast, not the building catalogue, decides.

IMPORTANT NOTE

Design the network first, then run routes within it

The order of operations in this lesson is not stylistic; it is the difference between planning and firefighting. The service promise and the demand pattern should set the network — where stock sits, where hubs sit, which lanes run direct — and daily routes then run within that structure. A network is chosen for the shipments of coming years; a route for this week's consignment.

When the order reverses, the symptoms are recognisable: stock regularly expedited from the wrong region to the right one; a hub used because it exists rather than because the flows justify it; direct runs invented overnight for lanes the structure cannot serve. Each rescue looks like dedication, and each is a signal that the daily routing is being asked to compensate for a network that no longer fits the flows inside it. This is how firefighting becomes institutionalised: permanently staffed by the operation's best people, who spend their skill rescuing deliveries instead of improving the system. The professional response is to treat recurring exceptions as network data — escalate the pattern into the network review, fix the structure, and let the routes run quietly inside it.

IMPORTANT NOTE

Empty legs are real money and real emissions

Every comparison in this lesson assumes vehicles earn their keep by carrying cargo. The empty leg breaks that assumption silently: it appears on no consignment file and carries no document, yet burns fuel, driver hours, and vehicle life at the full rate — and produces emissions with no freight to show for them. Backhaul planning is the discipline of making the return leg earn.

Three habits cover most of it. Check the return direction before committing to any lane: what physically moves back, and how much of it is ours to carry — supplier collections, empty pallets, reusable packaging, equipment repositioning? Build structural answers where flows repeat: the milk run of worked example three, reciprocal arrangements with partners trading in both directions, and network structures that end the vehicle's day where its next load begins. And price the empty leg honestly in route comparisons: a routing whose vehicles travel far empty to reach their loading point is dearer than its linehaul quote suggests, which is why door-to-door comparisons include positioning legs. An empty kilometre is pure cost and pure emissions; planning it away is among the cheapest improvements a transport operation can make.

IMPORTANT NOTE

A network designed for last year's demand is this year's constraint

Network decisions are investments, and investments age. Demand shifts — Lesson 13.1 showed how far demand moves with seasons, trends, and events. Costs shift: fuel, haulage rates, wages,

facility costs. Borders and corridors shift: Lesson 12.1 showed how procedures and infrastructure change the economics of whole regions. A structure that fitted the flows of three years ago can misfit today's: stock pooled where demand has thinned; a hub sitting on a lane that has grown dense enough to deserve direct lines; delivery radii stretched by customers the original design never imagined. The professional response is scheduled review, not crisis response. Watch the triggers: regional demand diverging materially from the pattern the network was designed around; transport or facility costs shifting the balance of the cost-versus-service trade-off; service performance degrading on particular lanes; border, corridor, or regulatory changes altering regional economics. Any one argues for re-running the network comparison with current data and with this lesson's total-cost honesty. Put the review triggers in the calendar as a matter of routine, the way audits are scheduled. Unexamined, a network does not stay neutral; it silently becomes the constraint the whole operation plans around.

Practice questions (Lesson Checks)

Practice Question 1

A planner must choose a routing for a recurring weekly consignment from Kumasi to a buyer in Ouagadougou. Routing A quotes GHS 40,000 door to door, crosses one border, and has arrived inside the buyer's delivery window in nine of the last ten trips. Routing B quotes GHS 37,500, crosses two borders, requires an intermediate transfer to a second carrier, and has missed the delivery window on about half its trips, with each miss triggering discount, waiting, and storage costs.

Which option best describes a sound choice?

1. Choose Routing B, because the lower quoted price is the only figure that actually appears in the transport budget.

2. Choose Routing A if its total expected door-to-door cost, including border waiting, handling risk, and the expected cost of missed windows, is lower, because reliability and handling are part of the route's real cost. ✓

3. Choose Routing B, because a routing with more border crossings always offers better regional coverage and therefore better service.

4. Split the weekly consignment between both routings so that at least half of each shipment is guaranteed to arrive on time.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

The lesson's central discipline is to compare routings door to door on total expected cost, with reliability and handling inside the comparison rather than outside it. Routing B's GHS 2,500 price advantage is only one line of the comparison. Its second border adds waiting days, its intermediate transfer adds a handling touch with damage and delay risk, and a window missed on roughly half the trips reliably triggers discount, waiting, and storage costs. Adding those lines, as worked example one did with similar numbers, routinely reverses a quote-based choice — and even where the totals came out level, the unpredictable routing would force the buyer to hold buffer stock against the bad case, a cost Answer 1 never counts.

Answer 3 invents a service benefit of extra borders that the lesson dismantled: every border is friction, waiting, and another document, not a feature. Answer 4 sounds cautious but is operationally incoherent: splitting a consignment across two arrival patterns means part of every shipment misses the window while both cost structures are paid in full.

The takeaway: the cheapest quote is a claim; the cheapest route is a total, and the total includes time, touches, and reliability.

Practice Question 2

A distributor serves six regional towns through a central hub. Flows between the towns are small and scattered, but one pair of towns an hour apart now exchanges two full truckloads of goods every day — all of it travelling into the hub, being unloaded and reloaded, and travelling back out to its destination.

Which statement best describes the sound network adjustment?

1. Route the dense lane's cargo direct between the two towns, while the hub continues to serve the thin, scattered flows that consolidate well. ✓

2. Keep the dense lane on the hub, because moving all flows through one place is always the cheapest arrangement regardless of volume.

3. Close the hub entirely and switch every flow to direct lines, because the dense lane has proved that direct delivery is always the superior structure.

4. Divide every consignment on the dense lane in half, so that the hub and a new direct line each carry part of the flow.

Correct answer: Answer 1

Explanation

The correct answer is Answer 1.

Structure follows flow. Thin, scattered flows between many towns consolidate efficiently through a hub — that is why the hub earned its place, and worked example two showed the

arithmetic. But a dense lane filling two complete trucks a day needs no consolidation: hubbing that cargo buys the worst of both structures, two spoke movements plus double handling where one full direct leg would do. The working pattern is hybrid — direct lines where flows are dense, hub-and-spoke where they are thin — with each lane reviewed as volumes change.

Answer 2 states the hub-always fallacy that the arithmetic disproves: a hub creates value by consolidating small flows into full vehicles, and a full load cannot be consolidated further, only detoured, delayed, and handled. Answer 3 overcorrects in the opposite direction; the other towns' thin flows still need the hub to fill their vehicles, and abandoning it would multiply part-load journeys. Answer 4 splits the flow for no operational reason, adding hub handling to half the cargo while paying for a direct line anyway.

The takeaway: review lanes by their actual flows and give each lane the structure its volume supports.

Practice Question 3

To shorten deliveries, a company replaces its single national warehouse with four regional distribution centres. A year later, total inventory is higher, some regions hold slow-moving stock while others stock out, and facility costs have risen.

Which option best explains what happened?

1. The company's demand forecasts must have been wrong, because the number of warehouses has no effect on inventory levels.

2. Fragmenting the network multiplied the buffers: each regional centre holds its own stock against its own region's demand, so total stock and fixed costs rise even as delivery distances fall — the cost-versus-service trade-off that network design must weigh. ✓

3. The regional centres are simply being managed badly, and centralising all inventory again would restore the previous performance without any other change.

4. The stockouts prove that regional distribution centres cannot work anywhere, and every network must remain fully centralised.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

This is the cost-versus-service trade-off of network design working exactly as the model predicts. More distribution centres shorten delivery distances and speed up response, but each added site fragments the company's inventory: every regional centre must hold its own buffer against its own region's uncertain demand, so the same nationwide protection now requires more total

stock. Fragmented buffers also explain the mixed symptoms — regions holding slow stock while others stock out — because stock in one region cannot serve demand in another. Fixed costs rise with the site count, since premises, staff, and systems come in sets.

Answer 1 is wrong on the mechanism: forecast quality matters, but the inventory rise is structural and would appear at any level of forecast accuracy. Answers 3 and 4 commit the mirror-image error the lesson warns against — treating one pole of the trade-off as universally correct. Centralising everything would cut inventory and fixed cost while lengthening every delivery; neither pole is correct. The design questions — the service promise, the market geography, the demand pattern — decide how many facilities the business should carry.

The takeaway: network structure sets inventory and cost behaviour, so change it only with the trade-off in view.

Fact-check appendix (audit notes — not for entry)

Definitions of Route Optimisation, Logistics Network, Distribution Network, Direct Delivery, Hub-and-Spoke Network, Vehicle Routing, Vehicle Capacity, Load Factor, Delivery Window, Route Constraint, Network Optimisation, Transportation Planning, Backhaul, and Last-Mile Delivery follow the lesson brief's key terms, defined at first use and used naturally in the blocks. — source: curriculum/lesson_briefs/module-13/lesson-13.2.md key terms.

Route planning as sequencing stops to minimise distance/time/fuel under capacity, time-window, and driver-hours constraints (the vehicle-routing problem family); constraints taught in Key Concept on routing. — source: research sheet logistics-scm-frameworks.md section 6.

Cost trade-off between transport and inventory (slower cheaper modes raise inventory and stockout costs; total-cost principle); taught as the total-system discipline. — source: research sheet logistics-scm-frameworks.md section 6.

Consolidation concepts: FTL dedicates a vehicle to one shipper (faster, direct, less handling); LTL combines several shippers' freight in one vehicle (cheaper per unit, more handling, longer transit); groupage is the consolidation term in freight forwarding; LTL carriers typically run hub-and-spoke terminal networks. — source: research sheet logistics-scm-frameworks.md section 6.

Milk run: fixed, repeated circular route with multiple pickup/drop-off stops, consolidating small loads. — source: research sheet logistics-scm-frameworks.md section 6.

Hub-and-spoke consolidates flows through central hubs (fewer connections, longer transit) versus point-to-point (faster, needs volume per pair): standard network-design convention. — source: research sheet logistics-scm-frameworks.md section 6.

Multimodal (at least two modes, single contract, one operator responsible end to end) versus intermodal (goods in one and the same loading unit across modes, often separate contracts): taught per the sheet's distinction; no claim that intermodal means single contract. — source: research sheet logistics-scm-frameworks.md section 6 and DO NOT CLAIM item 2.

Empty running (travelling without paying cargo) is a major cost and emissions inefficiency, cut via backhaul matching and network design; backhaul is the return leg. Taught qualitatively; no percentages claimed. — source: research sheet logistics-scm-frameworks.md section 6 and DO NOT CLAIM list.

Load-factor improvement and empty-run reduction as recognised emissions levers; capacity utilisation as a core efficiency lever for cost and emissions; a moving vehicle burns fuel regardless of fullness (qualitative). — source: research sheet logistics-scm-frameworks.md sections 6, 7, and 12.

Network design: number/location of distribution centres is a cost-versus-service trade-off — more/deeper networks cut delivery time and outbound transport but add facility and inventory cost; centralised networks do the reverse; inventory fragmentation (each site holding its own buffer) taught from this trade-off. — source: research sheet logistics-scm-frameworks.md section 8.

Postponement (delayed differentiation): keep the product generic and defer final configuration until real demand is known; suited to high demand uncertainty. — source: research sheet logistics-scm-frameworks.md section 8.

Cross-docking: goods move from receiving dock to shipping dock with little or no storage. — source: research sheet logistics-scm-frameworks.md section 4.

TMS scope: plans and executes transport — mode/route selection, carrier selection and rating, tendering, tracking, freight audit/settlement. No vendor names, no ROI or savings percentages anywhere in the lesson. — source: research sheet logistics-scm-frameworks.md section 9 and DO NOT CLAIM item 10.

Routing/scheduling software and network-design tools taught as computing and comparing options on the inputs given (garbage in, garbage out); software claims limited to functions. — source: research sheet logistics-scm-frameworks.md sections 6 and 9.

Resilience via alternative sites/structures (a second distribution centre adding resilience) taught as a concept. — source: research sheet logistics-scm-frameworks.md section 11 (resilience strategies).

Geography of worked example one: Ouagadougou is connected by paved highways to Accra and to Lomé (paved Accra–Ouagadougou axis exists); Burkina Faso is landlocked and reaches the sea through coastal neighbours' gateways. Border crossings are described generically ("the northern border crossing with Burkina Faso", "the coastal border into Togo"): Aflao, Paga, and Elubo are marked Not Verified as named border posts in the research sheet, so no border post is named anywhere in the lesson. — source: research sheet ghana-trade-customs.md section 8 including its Not Verified note; consistent with the draft of lesson 12.1.

The West African regional road transit arrangement (one document, seal, guarantee; sealed loads not reloading en route) is referenced only as "the regional transit arrangement met in Lesson 12.1" with no operational specifics re-taught. — source: research sheet trade-transport-standards.md section 12; draft lesson-12.1.

Vehicle weight limits and weighbridges taught generically as the legal side of vehicle capacity; no country-specific limits stated. — source: general business knowledge; corridor control points per draft lesson-12.1, WRITING_SPEC.md rule 5.6.

All figures in all four worked examples (quotes, penalties, pallet counts, load factors, buffer stocks, daily costs) are round illustrative teaching numbers, labelled as such in each example; no duty rates, fees, or statutory periods are stated anywhere. — source: WRITING_SPEC.md rule 5.1.

Cross-references by name and number, no re-teaching: Lesson 3.2 Modes of Transport: Sea, Air, Road, Rail, and Multimodal (mode choice, multimodal/intermodal); Lesson 3.3 Cargo Types, Packaging, and Handling (handling risk); Lesson 5.3 Shipping Instructions and Cargo Consolidation (consolidation); Lesson 7.4 Shipping Charges, Freight Rates, and Surcharges (charge-build scope); Lesson 9.2 (inland depots); Lesson 10.1 Components of Total Logistics Cost (total-cost lens); Lesson 12.1 International Transport Corridors (border friction, corridors, transit); Lesson 13.1 Demand Forecasting and Supply Planning (previous lesson, demand pattern); Lesson 13.3 Inventory Optimisation and Service Levels (next lesson, forward reference). — source: WRITING_SPEC.md section 7; lesson briefs for modules 3, 5, 7, 9, 10, 12, and 13.

Lesson 13.3 — Inventory Optimisation and Service Levels

Summary

This lesson explores how organisations determine appropriate inventory quantities while balancing product availability, customer service, and operating costs. Learners will examine inventory classification, reorder points, safety stock, economic order quantity, and service-level targets. The lesson also explains how demand variability and replenishment lead times influence inventory decisions and how excessive or insufficient stock affects supply chain performance.

Learning objectives

- Define inventory optimisation and explain its purpose.
- Explain the relationship between inventory availability and customer service.
- Identify the major costs associated with holding and replenishing inventory.
- Describe the purpose of safety stock and reorder points.
- Explain the principles of economic order quantity.
- Distinguish between cycle stock and safety stock.
- Describe how ABC inventory classification supports stock management.
- Explain how demand variability and lead time influence inventory requirements.
- Calculate basic reorder points and inventory measures using supplied information.
- Evaluate inventory policies based on cost, stock availability, and service-level objectives.

Key terms

Inventory Optimisation — The process of determining inventory policies and quantities that balance cost, availability, and service requirements.

Service Level — A defined measure of the ability to satisfy customer demand or meet specified service expectations.

Safety Stock — Additional inventory maintained to reduce the risk of shortages caused by uncertainty in demand or replenishment.

Cycle Stock — Inventory held to satisfy expected demand between replenishment orders.

Reorder Point — The inventory position at which a replenishment order should be initiated under a defined policy.

Economic Order Quantity (EOQ) — An inventory model used to estimate an order quantity that minimises specified ordering and holding costs under its assumptions.

ABC Analysis — An inventory classification technique that groups items according to their relative importance, commonly based on annual consumption value.

Inventory Holding Cost — The cost associated with maintaining stock over time.

Ordering Cost — The expense incurred in placing and processing replenishment orders.

Stockout — A situation in which an inventory item is unavailable when required.

Fill Rate — The proportion of demand fulfilled immediately from available inventory under a defined measurement method.

Inventory Turnover — A measure of how frequently inventory is sold or consumed and replaced during a specified period.

Replenishment Lead Time — The period between initiating a replenishment order and receiving the goods.

Inventory Position — A measure of stock availability that commonly includes on-hand inventory and outstanding orders, adjusted for backorders.

Teaching content

KEY CONCEPT

Inventory optimisation: the strategy above the mechanics

Lesson 8.3, Inventory Control, FIFO, and FEFO, taught the working mechanics of stock: how to issue it (FIFO and FEFO), how to record it, when to reorder (the reorder point), how much to order (the economic order quantity), and how to rank items for control effort (ABC analysis). This lesson moves one level up, to the questions those mechanics exist to serve. What availability should the operation promise? How much of the organisation's money should sit on shelves to keep that promise? And which stock policies turn that money into service most efficiently? Answering them is the work of inventory optimisation, the process of determining inventory policies and quantities that balance cost, availability, and service requirements.

Three habits of mind define the optimiser.

1. Inventory is invested cash. Every unit on a shelf is working capital that cannot be spent elsewhere until the unit sells or is used. A distributor holding stock worth two million cedis is not merely storing goods; the business has placed two million cedis of its capital inside its own warehouses, and optimisation asks what service that placement buys.

2. The purpose is balance, not minimisation. The cheapest possible inventory policy is no inventory at all, and it is also a guarantee of stockouts, lost customers, and emergency purchases. The most generous policy inverts the failure. Optimisation rejects both extremes: it searches for the policy at which the total cost of holding stock and of failing to hold it is lowest.

3. Policies are set, not inherited. Reorder points, safety stocks, and order quantities are decisions somebody makes, explicitly or by default. The optimiser makes them explicitly, from demand, lead times, and costs, and revisits them when those inputs move.

The module has already assembled the inputs. Lesson 13.1 supplied the demand view, including forecast error; Lesson 13.2 supplied the network view, the number and location of stocking points. This lesson converts both into stock policy, and measures the result in service and in cash.

KEY CONCEPT

Service level: a designed availability target, not an outcome

The service level is a defined measure of the ability to satisfy customer demand or meet specified service expectations, and the single most important fact about it is that it is a policy choice. An organisation decides, in advance, the availability it will design its stock to deliver, in the same way an engineer decides the load a bridge is built to carry. Two readings of the term are common, and the professional habit is to be explicit about which one is in use.

1. Availability target. The cycle service level is the target probability of passing through a replenishment cycle without a stockout: a 95 per cent cycle service level means that 95 replenishment cycles in every 100 are designed to end with demand still covered when the next order arrives. This is a property the organisation designs into its safety stock.

2. Fulfilment measure. The fill rate is the proportion of demand fulfilled immediately from available inventory under a defined measurement method. It reports what actually happened to customers. Its stricter cousin, the on-time in-full measure examined in Lesson 13.4, Logistics Performance Indicators and KPIs, scores an order as a success only if it arrived both on time and complete; and when a promise slips despite the design, telling the customer honestly is its own discipline, met in Lesson 5.4, Shipment Tracking and Customer Communication.

The distinction matters because a designed target and a measured outcome are different kinds of statement. A target without measurement is a hope; a measurement without a target is an accident that has been scored. Optimisation uses both: the target to size the buffers, the fulfilment measures to check whether the design delivered.

Service levels also differ by product and by customer, and legitimately so. A hospital pharmacy and a confectionery wholesaler should not carry the same availability promise, and neither should a contracted key account and casual walk-in demand. The promise is differentiated, deliberately, and the next blocks give that differentiation its logic.

KEY CONCEPT

The costs that compete: holding, ordering, and stockouts

Optimisation begins with a complete cost picture, because every inventory policy is a choice among costs. Three families compete.

1. Inventory holding cost. The cost associated with maintaining stock over time: the capital tied up in the goods, the storage space they occupy, insurance, shrinkage and damage, and the risk of obsolescence as stock ages. Lesson 10.1, Components of Total Logistics Cost, made the

uncomfortable point that carrying cost is real money even though no invoice names it; it surfaces as interest, rent, write-offs, and lost liquidity. Holding cost grows with the quantity held and the time it is held.

2. Ordering cost. The expense incurred in placing and processing replenishment orders: the paperwork and approval behind each purchase order, receiving and inspection at the warehouse, invoice matching, and the transport and administration that accompany every consignment. Its shape matters: much of it is charged per order rather than per unit, so frequent small orders cost more in total than fewer large ones for the same annual demand.

3. Stockout cost. A stockout is a situation in which an inventory item is unavailable when required, and its costs arrive from several directions at once: the margin on the sale that could not be made, the expediting cost of emergency replacements, the customer who quietly moves to a competitor, and the contractual or reputational damage of an unkept commitment. Lesson 10.1 grouped these with the costs of failure, the family nobody budgets but everybody eventually pays.

The strategic tension is now visible. Holding more stock attacks stockout costs and inflates holding costs. Ordering in larger, rarer quantities attacks ordering costs and inflates holding costs. No policy escapes all three; a good policy chooses deliberately among them.

KEY CONCEPT

The optimisation problem: two costs pulled against each other

State the problem honestly. Raising the service level requires more safety stock; more safety stock raises the holding cost. Cutting the holding cost means thinner buffers; thinner buffers raise the frequency and severity of stockouts, and stockouts cost money too. The optimisation problem is therefore not to minimise inventory and not to maximise availability, but to find the policy at which the sum of the two competing costs is smallest. In principle, the optimum sits where the money saved by one more step of availability equals the money that step costs to carry.

Everything depends on pricing the second cost honestly, so the optimiser estimates the cost of a stockout before choosing a target. The working approach is deliberately practical.

1. Size the event. For a typical stockout of the item, estimate how many units of demand would go unmet before the next replenishment arrives.

2. Price the margin at risk. Multiply those units by the gross margin each would have earned. This gross-margin-at-risk figure is the irreducible core of the stockout cost.

3. Add the consequences. Add the expediting cost of an emergency replacement and, conservatively, the value of harmed customer relationships or contractual penalties. Some consequences, such as patient harm from a missing medicine, cannot be honestly priced in

money; the professional response is to treat the monetary estimate as a floor rather than a ceiling, and to let that item earn a higher protection level.

The estimation is rough by design, and that is acceptable: its job is to rank policies against each other, not to produce a figure for the annual report. A rough stockout cost that is written down and argued over is worth more than a precise silence, because the commonest and most expensive error in inventory management is simply pretending the cost is zero.

KEY CONCEPT

Cycle stock and safety stock: two buffers, different jobs

Average inventory answers two different questions, and the optimiser keeps the answers apart.

Cycle stock is inventory held to satisfy expected demand between replenishment orders. It exists because of ordering economics: ordering cost is charged per order, so an organisation orders in batches even though demand arrives continuously, and each batch drains away as customers buy. Cycle stock is a bet that demand will behave as expected, and its size is an order-quantity decision. The classic instrument is the economic order quantity taught in Lesson 8.3: a model that estimates an order quantity minimising the sum of ordering and holding cost under strong simplifying assumptions, including constant demand, full delivery at a single stroke, fixed order cost, and no quantity discounts or stockouts. The principles survive into advanced practice: order size is a balance between ordering cost and holding cost, and the assumptions are conditions to check, not decorations to ignore.

Safety stock is additional inventory maintained to reduce the risk of shortages caused by uncertainty in demand or replenishment. It exists because Lesson 13.1's forecasts are wrong by some amount and lead times wobble. Safety stock is not a bet on demand behaving; it is the price paid for demand being unsteady.

The reorder point is the inventory position at which a replenishment order should be initiated under a defined policy. It triggers on the inventory position, a measure of stock availability that commonly includes on-hand inventory plus outstanding orders, adjusted for backorders, so that goods already ordered and on their way are counted where they should be. Its construction is the one formula carried over from Lesson 8.3: expected demand during the replenishment lead time, plus safety stock.

Strategically: cycle stock is governed by order economics, safety stock by uncertainty, and the reorder point is where the two meet the calendar.

KEY CONCEPT

What swells the buffer: demand variability and replenishment lead time

Two forces determine how much protection an item needs, and both were met earlier in the programme; this block states their inventory consequences precisely.

1. Demand variability. The noisier the demand during the exposure window, the more safety stock the same service promise requires. An item selling a steady 30 units every day can be protected with a thin buffer, because a bad day is barely worse than an average one. An item selling 30 units on average but ranging from 5 to 90 needs a thick one, because the buffer must absorb the spread, not the average. This is why Lesson 13.1's work on forecast error matters twice: a better forecast improves the plan directly and shrinks the buffer every policy must carry.

2. Replenishment lead time. The period between initiating a replenishment order and receiving the goods. Its components, from the programme's earlier lessons, are order processing, preparation and pick-pack, transport and handling, and, across borders, customs clearance. A longer lead time extends the exposure window, so more expected demand must be covered and more can go wrong inside the window. A less reliable lead time is worse than a merely long one: variability in the lead time itself loads additional uncertainty onto the same buffer.

The consequence is the hinge of this lesson. The buffer is sized against what can go wrong during exposure. Shrink the exposure, in length or in variability, and the same service promise can be kept with a smaller buffer. That observation converts inventory optimisation from a choice between overstocking and understocking into a design discipline with levers, and the next blocks take up those levers in turn.

KEY CONCEPT

Not every product deserves the same promise: segmenting service levels

A single service level applied to a whole assortment is quietly wasteful in both directions: it overspends on items whose stockouts hardly matter and under-protects the items whose stockouts are expensive or dangerous. The optimiser's alternative is segmentation, setting different availability targets for different products and customers, each justified by the balance at the heart of the definition of inventory optimisation introduced earlier: the cost of protection against the cost of failure.

The product dimension. A life-critical medicine and a convenience good pass through the same mechanics but face different consequences. When the medicine is missing, the cost of the stockout is extreme, and not only in money, so the organisation designs availability near the top of the practical range. When the multivitamin is missing, the customer buys another brand or returns next week, so a more modest target is the honest optimum. Neither product is being treated unfairly; each is priced according to what its absence costs.

The customer dimension. A contracted key account with a measured availability commitment and a casual walk-in buyer justify different policies for the same item, and the contract, not habit, should decide.

The instrument for doing this systematically is ABC analysis, which Lesson 8.3 taught as a classification that groups items according to their relative importance, commonly based on

annual consumption value: a few A items carry most of the value, and a long tail of C items carries little. At the strategic layer, ABC becomes the skeleton of the whole stock policy: A items earn tight service targets, frequent review, and first claim on management attention, because that is where the money and the service concentrate; C items earn cheap, simple policies, because elaborate protection of trivial value is optimisation inverted. Classification supports stock management precisely by making differentiated treatment systematic instead of personal.

KEY CONCEPT

Lever 1: shorten and stabilise the lead time

The first optimisation lever does not touch the warehouse at all. Because the buffer is sized against what can go wrong during the replenishment lead time, anything that shortens or steadies that window shrinks the stock needed for the same promise.

Shortening works through speed. The programme's freight lessons, Lesson 7.1, Ocean Freight Operations and Vessel Schedules, and Lesson 7.2, Air Cargo Operations and Cargo Acceptance, showed the two speeds available on international lanes, and Lesson 10.1's air-versus-sea comparison showed the total-cost way to read them: air costs more per consignment and may cost less in total once the inventory it eliminates is priced. Moving a critical item from a six-week ocean pipeline to a shorter air one converts weeks of exposure into days, and weeks of exposure are precisely what the safety stock was bought to cover.

Stabilising works through reliability, and it is usually the cheaper half of the lever. Dependable supplier performance, prompt order processing, fewer customs surprises through accurate documentation, and carriers that keep their schedules all narrow the spread of the lead time without buying speed. Since the buffer must absorb the spread, narrowing it is service delivered by organisation rather than by stock.

The caution is the one Lesson 10.1 taught: freight savings and inventory savings live in different ledgers, so a mode change that looks expensive to transport can look cheap to the business, and one that looks cheap to transport can quietly hold its cost in safety stock on a shelf. The optimiser compares the whole, transport and inventory together, before declaring a lead-time change an improvement.

KEY CONCEPT

Lever 2: reduce the variability you are exposed to

The second lever attacks the other input to the buffer: variability itself. Three instruments do this work.

1. Better forecasting. Lesson 13.1 established that forecast error is measured and managed, and its inventory price is now visible: every unit of typical error must be carried as safety stock somewhere. Narrow the error, with better data, better methods, and the sales and operations planning conversation, and every buffer downstream can shrink at unchanged service.

2. Postponement. The technique of keeping the product generic and deferring final configuration, such as dyeing, labelling, local packaging, or final assembly, until real demand is known. Its inventory power is structural: uncertainty attaches to the small pool of generic stock rather than to the large family of finished variants, so the buffer is carried once instead of many times. It suits products whose final form can be decided late and whose demand is uncertain, and it is the same idea Lesson 13.1 met from the planning side as delayed differentiation.

3. Calmer demand signals. Lesson 13.1's bullwhip effect showed how order swings amplify upstream through forecast updating, order batching, price fluctuations, and shortage gaming. A stockholder who sees distorted orders is forced to buffer against phantoms; shared point-of-sale data, smaller and more frequent orders, and steady pricing shrink the swings and therefore the buffers built on them.

Under all three sits a floor: record discipline. Lesson 8.4, Warehouse Documentation and Stock Reconciliation, made the operational case for accurate stock records; the strategic case is that every parameter in this lesson, safety stock, reorder point, service target, is computed from records, and a policy computed on wrong records protects a fiction while the real stock quietly does something else.

KEY CONCEPT

Lever 3: pool inventory across locations

The third lever is geographic. When the same item is stocked at many locations, each location carries its own safety stock against its own local surprises, and the totals mount: five stores each hedging their own uncertainty hold five separate hedges. When the item is pooled into fewer locations, the same total demand is served with less total buffer, because local swings partly cancel: the week one region runs hot, another often runs cool, so the pooled demand is steadier than the sum of its parts. Pooling converts separate, larger hedges into one smaller hedge for the same coverage, and the worked example later in this lesson puts illustrative arithmetic on the effect.

Pooling is the inventory face of the decision Lesson 13.2, Transport Route and Network Optimisation, analysed from the transport side. The trade-off there is symmetric: more stocking points in the network cut delivery time and outbound transport cost but add facility and inventory cost, while a centralised network does the reverse. The optimiser reads the network decision as an inventory decision too: every stocking point added is another buffer purchased, and every one consolidated is buffer released.

The lever has honest limits, and optimisation respects them. Gains shrink when local demands rise and fall together, because swings that move together do not cancel. Pooling lengthens the response to distant customers, which is a service cost paid in delivery time rather than in stock. And some stock must sit close to demand regardless: fast movers, and items whose absence stops a customer's work within hours. The question is therefore not whether to centralise, but

which items pool well and which must live near the customer, decided item by item along the ABC lines drawn earlier.

KEY CONCEPT

Measuring inventory performance: turnover, days of supply, and the cash view

A policy is not performing unless it can be scored, and inventory performance has two standard instruments.

1. Inventory turnover. A measure of how frequently inventory is sold or consumed and replaced during a specified period, computed as the cost of goods sold divided by average inventory for the period. A turnover of six says the operation sells and replaces its average stock six times a year: the same stock funds six buying seasons inside twelve months.

2. Days of supply. Average inventory divided by average daily usage, answering the plainer question: if demand continued as usual, how many days would today's shelves last? When usage aligns with the cost figure, days of supply approximates 365 divided by turnover, and the optimiser states which denominator is in use, because the two are reported differently in different businesses.

Both feed the working-capital lens this lesson has pressed throughout: inventory is cash on shelves. A reduction in days of supply at unchanged service is cash released, and it is one of the few improvements a finance director and a customer can applaud together.

The optimiser's red flag is aged and obsolete stock: items that no longer turn, expiry-dated goods approaching their dates, models superseded, spares for machines already retired. Aged stock is not merely expensive; it is diagnostic. Stock that nobody buys was ordered under a policy nobody revised, so the warehouse floor is where stale assumptions resurface as write-offs. Reviewing turnover and days of supply item by item finds the flag while it is still a flag.

One caution guards the whole block: turnover alone is not a verdict. A low figure can be negligence, or it can be a deliberate buffer for a critical spare with a long lead time, and the difference lies in the item's role, as the final worked example demonstrates.

KEY CONCEPT

The review rhythm: optimisation is a cycle, not a project

Every parameter this lesson has discussed, the service level targets, the safety stocks, the reorder points, was computed from assumptions: a demand rate and its variability, a lead time and its reliability, a set of costs. None of those assumptions holds still. Demand drifts and seasons shift, and Lesson 13.1's forecast error guarantees the plan will be wrong by some amount; suppliers change performance and lead times move; margins, freight rates, and the cost of capital all wander. An inventory policy that was optimal when set is silently drifting away from optimal ever after, which is why optimisation is a cycle, not a project.

The rhythm has three beats.

1. Set. Choose service level targets by segment, size safety stocks and reorder points from demand, lead-time, and cost data, and write the policies down so they can be examined.

2. Measure. Score the design against reality: stockout and fill-rate history against the targets, turnover and days of supply against intention, aged stock against a zero-tolerance line.

3. Revise. When measurement or changed assumptions show drift, recompute the parameters. A target that keeps missing, a buffer that never once triggered, an item whose lead time halved, each is a signal to reset, not a curiosity.

The cadence is part formal, part event-driven: scheduled reviews, often folded into the sales and operations planning cycle Lesson 13.1 described, plus immediate reviews when a supplier changes, a product launches or retires, or a disruption moves lead times. Lesson 8.3 closed on the note that the formulas are servants, not masters; the strategic version of that note is that the parameters are servants too. They serve the balance between cost and service, and they keep their jobs only while the balance holds.

WORKED EXAMPLE

Two products, two promises: setting service levels at an Accra distributor

A health-and-wellness distributor in Accra carries two items through the same warehouse and the same ordering routine: an essential injectable medicine supplied to hospitals, and a mass-market multivitamin supplement sold through pharmacies and shops. Both have annual demand of about 12,000 units, both are ordered in batches of 1,000, so both pass through about 12 replenishment cycles a year, and both face a lead time of 10 days, about 33 units of expected demand per day. All figures in this example are illustrative for teaching purposes.

Step 1: Price a stockout for each item. A typical medicine stockout leaves roughly 150 units of hospital demand unmet. At a gross margin of GHS 20 per unit, the margin at risk is GHS 3,000; expediting a replacement costs about GHS 1,000; and harm to hospital relationships is valued conservatively at another GHS 1,000, since a missing essential medicine can also harm patients in ways money does not capture. The working estimate is GHS 5,000 per event, treated as a floor. A typical multivitamin stockout strands about 400 packs at a margin of GHS 3 each, GHS 1,200, and little else: customers substitute or return, and no expedited freight is justified.

Step 2: Compare a 90 per cent and a 99 per cent target for the medicine. At 90 per cent, each of the 12 annual cycles carries a 10 per cent chance of failure, so 1.2 stockout events are expected: 1.2 times GHS 5,000, or GHS 6,000 a year. At 99 per cent, expected events fall to 0.12, about GHS 600, but protection grows disproportionately: safety stock rises from 250 to 600 units, moving the reorder point from 330 plus 250 (580) to 330 plus 600 (930). The extra 350 units cost an assumed GHS 6 per unit-year, GHS 2,100, against about GHS 5,400 of expected stockout cost avoided; the raise pays for itself before counting the unpriced patient consequences.

Step 3: Run the same comparison for the multivitamin. At 90 per cent, expected stockout cost is 1.2 times GHS 1,200, or GHS 1,440 a year; at 99 per cent it falls to about GHS 144, saving roughly GHS 1,300. But the extra 400-pack buffer costs GHS 5 per pack-year, GHS 2,000, so the raise costs more than it saves: the lower target is not neglect but the honest optimum for this item.

What does this example demonstrate? The same mechanics produced opposite policies, because the two items price failure differently.

Key lesson: a service level is chosen item by item, weighing the cost of protection against the cost of failure; for a life-critical product, failure sets the policy.

WORKED EXAMPLE

One store or two: the pooling effect between Tema and Kumasi

A food wholesaler sells one steady line of tinned tomatoes through two regional stores, one in Tema serving the south and one in Kumasi serving the north. Each store averages 600 cartons of demand a month. Demand wobbles: a month can run some 200 cartons above or below average at either store, and the swings do not move together, because a strong month in the south is often a quiet month in the north. Each store therefore holds a safety stock of 200 cartons sized against its own swings, 400 cartons of buffer in total. All figures in this example are illustrative for teaching purposes.

A consultant proposes consolidating the buffer: hold this line's safety stock at a single central store in Tema serving both regions. The reasoning runs in four steps.

Step 1: Combine the exposure. The central store faces 1,200 cartons of average monthly demand, but not the swings of either single store. Because the two regions' good and bad months partly offset one another, the combined demand is steadier than either half: the pooled swing is smaller than the sum of the swings.

Step 2: Size the pooled buffer. The buffer exists to absorb variability, and pooled variability is proportionally smaller, so the same coverage can be designed with less total stock. In this illustrative design, 280 cartons at the central store replace the 400 held across the two sites.

Step 3: Bank the release. The 120 released cartons, at an illustrative value of GHS 150 each, return about GHS 18,000 of working capital at once, and they no longer cost the assumed GHS 7.50 per carton-year to carry, about GHS 900 a year saved, with designed availability unchanged.

Step 4: Pay the price of pooling, knowingly. Northern customers are now served from Tema, so their replenishment rides the Tema to Kumasi leg: delivery to the north becomes slower, or its outbound transport dearer, which is exactly the facility-and-inventory cost against delivery-time-and-transport-cost trade-off that Lesson 13.2 placed on the other side of the same decision. And the gains have conditions: if the two regions' demands rose and fell together, as they would under a shared promotion or a shared festive season, the swings would reinforce rather than cancel, and the pooled buffer would approach the sum it replaced.

What does this example demonstrate? Pooling serves the same total demand with a smaller total hedge, because independent variability partly cancels when demands share one pool.

Key lesson: the pooled buffer buys the same availability for less money, and the currency it pays in is delivery speed to the periphery.

WORKED EXAMPLE

Two SKUs in Rotterdam: turnover, days of supply, and the role of the item

A marine spare-parts distributor in Rotterdam supplies engine components to shipping companies. A finance review of its inventory produces two striking figures, and the skill this example teaches is reading them. All figures in this example are illustrative for teaching purposes.

Step 1: Measure SKU P, a fast-moving line of engine filters. Annual usage at cost is 480,000 US dollars, and average inventory is 40,000 US dollars. Inventory turnover is the cost of goods sold divided by average inventory: 480,000 divided by 40,000, which is 12 times a year. Days of supply is average inventory divided by average daily usage: daily usage at cost is 480,000 divided by 365, about 1,315 dollars a day, so the stock on hand represents 40,000 divided by about 1,315, or roughly 30 days. The line turns about monthly, tidy and disciplined.

Step 2: Measure SKU Q, a specialised pump impeller for one family of engines. Annual usage at cost is 60,000 US dollars, and average inventory is 45,000. Turnover is 60,000 divided by 45,000, about 1.3 times a year. Daily usage at cost is about 164 dollars, so days of supply is 45,000 divided by about 164, roughly 274 days. On a turnover league table, SKU Q looks like a disaster: a part that sits on the shelf for the better part of a year.

Step 3: Read the figures against each item's role. SKU Q is a critical spare whose maker is sixteen weeks away by the replenishment lead time, and whose absence can hold a customer's vessel in port. The 274 days on the shelf are not drift; they are the policy, a buffer bought deliberately because the lead time is long and failure is expensive. Judged by turnover alone, the distributor would starve exactly the item its customers depend on. SKU P, by contrast, is replenished locally within days; if its days of supply crept from 30 toward 60, the review would rightly ask what cash was doing on those shelves.

Step 4: Keep the red flag in view. Role justifies SKU Q's buffer, but it does not freeze it: if the engine family is retired, or the maker opens a nearer depot, the policy must be recomputed. An impeller bought for engines no longer sailing is precisely the aged-stock flag this lesson warned about, surfacing eventually as a write-off.

What does this example demonstrate? Turnover and days of supply locate the questions; they do not answer them.

Key lesson: inventory measures are read against the item's role, its lead time, and its criticality, never against a universal league table.

IMPORTANT NOTE

One hundred per cent service level is not a strategy

The most seductive number in inventory management is 100 per cent, and the most expensive. Safety stock does not grow in neat steps as the target rises; protection grows disproportionately as the target approaches completeness, because the buffer must stretch to cover the increasingly extreme tail of what demand and lead time might do. A 90 per cent target protects against the plausible; a 99.9 per cent target protects against the freak; promising to cover every conceivable outcome means carrying worst-case stock on every item, every day. The carrying cost of that promise becomes crushing for practical purposes: cash converted into shelves, insurance, and, for dated goods, eventual write-offs.

It is also a promise no stock policy can keep. Suppliers fail, quality rejections happen, and the freak outcome the buffer never anticipated occasionally arrives anyway. The professional position is not the perfect number but the chosen number: a target set deliberately for each segment of the assortment, with its residual risk named and accepted by the people accountable for it. An organisation that announces 100 per cent has not made a policy decision; it has declined to make one, and it will discover its real target the first time the cash runs out or the warehouse fills.

IMPORTANT NOTE

A service level without a stock policy is marketing

Announcing a service level changes nothing by itself. A 99 per cent availability promise is delivered, or not, by the machinery beneath it: safety stock actually purchased, reorder points actually honoured, order quantities actually reviewed, lead times actually managed, records actually accurate. If the buffers were never resized to support the new number, the announcement has merely raised the standard against which the operation will now visibly fail. That is the first half of the warning: a service level without the stock policy to deliver it is marketing.

The second half is the mirror image. A stock policy without a stated service level drifts just as badly, because nobody can say what the buffers are for. Every stockout then looks like negligence, every surplus looks like prudence, and arguments about stock are settled by seniority instead of by the balance between cost and failure this lesson taught. The target and the parameters are two halves of one instrument, and the measurement discipline, fill-rate history, turnover, days of supply, is the third. Set the target, size the policy to it, measure the result, and revise: the cycle only works when all three are present, and an organisation holding any two of them is decorating a gap.

Practice questions (Lesson Checks)

Practice Question 1

The managing director of a distribution company tells the quarterly review: "From next month, we will hold a 100 per cent service level on every product. I want stockouts to become impossible."

Which statement best evaluates this instruction?

1. The instruction is sound provided the company enlarges its warehouses before raising the service levels.

2. The instruction mistakes a policy decision for its abolition: protection grows disproportionately as the target nears completeness, and covering every conceivable outcome carries carrying costs that soon exceed any stockout cost, so professional practice sets a chosen target for each segment with its residual risk named and accepted. ✓

3. The instruction is achievable because service levels are controlled by the stock the company already holds, and existing buffers were sized generously enough.

4. The instruction will succeed because stockouts are caused mainly by suppliers, so a 100 per cent internal target removes them.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

A service level is a policy choice: the availability the stock system is designed to deliver, balanced against the cost of providing it. As the target approaches completeness, safety stock grows disproportionately, because the buffer must stretch over the increasingly extreme tail of what demand and lead time might do. Carrying worst-case stock on every item converts cash into shelves, insurance, and eventual write-offs, a cost that for practical purposes outgrows the stockout cost it prevents. The professional alternative is a target set deliberately for each segment of the assortment, with the residual risk named and accepted.

Answer 1 accepts the instruction and answers only the storage question, missing that the cost problem lies in the stock itself. Answer 3 is wrong on both counts: existing buffers were sized to the old targets, and no realistic buffer makes stockouts impossible, since suppliers fail and extreme outcomes arrive. Answer 4 blames suppliers alone, but internal buffer sizing is exactly what determines whether a supplier's delay turns into a customer-facing stockout.

The takeaway: 100 per cent is not a strategy; it is a refusal to balance costs, and the balance is the strategy.

Practice Question 2

A wholesaler serves the north of the country from a Kumasi store and the south from a Tema store. Each store holds its own safety stock against its own local demand swings, and the swings at the two sites do not move together. A consultant proposes holding this line's safety stock in one central store at Tema instead.

Which statement best describes the inventory effect of the proposal?

1. The central store must hold the sum of the two regional safety stocks, because total demand is unchanged, so no stock is released.

2. The pooled safety stock can be smaller than the combined regional buffers, because independent local swings partly cancel when demands share one pool, though replenishment to the more distant region becomes slower or costlier to transport. ✓

3. Pooling eliminates stockouts entirely, because one large store cannot run short when two smaller stores could.

4. Pooling has no effect on total stock, because safety stock depends only on the item's unit value and not on where it is held.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

Safety stock exists to absorb variability, and pooled demand is steadier than the sum of its parts: when the two regions' good and bad months partly offset, the combined swing is smaller than the swings added together, so the same designed availability can be achieved with a smaller total buffer. The lesson's worked example put illustrative arithmetic on exactly this effect. The price of pooling is paid in service to the periphery: distant customers now wait for the longer delivery leg, the same facility-and-inventory against delivery-time-and-transport trade-off the network lesson placed on the other side of the decision.

Answer 1 commits the commonest pooling error: adding the buffers as if variability added too, when the cancelling of swings is the entire source of the gain. Answer 3 overclaims: pooling shrinks the expected shortage, but no stock level eliminates it, and correlated swings such as a shared festive season erode the gain. Answer 4 denies the effect altogether; safety stock is sized against variability and lead time, both of which change when stock is pooled.

The takeaway: pooling buys the same availability with less total hedge and pays in delivery speed.

Practice Question 3

A distributor's stock review shows two items. A consumable filter turns over 12 times a year, about 30 days of supply. A specialised pump impeller for one engine family turns over about 1.3 times a year, about 274 days of supply, and is ordered from a maker sixteen weeks away. A manager proposes cutting the impeller stock until its turnover matches the filter's.

Which statement best evaluates the proposal?

1. The proposal is sound, because a higher turnover is always better and any item with low turnover is over-stocked by definition.

2. The proposal is unsound, because the impeller's high days of supply is a deliberate buffer against a long lead time and expensive failure; the measures locate questions rather than answer them, and judgement depends on the item's role, lead time, and criticality. ✓

3. The proposal is unsound because turnover cannot be measured for spare parts, only for consumables.

4. The proposal is sound, because both items are selling steadily and therefore their stock positions must be equally healthy.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

Inventory turnover, the cost of goods sold divided by average inventory, and days of supply, average inventory divided by average daily usage, are diagnostic instruments: they show where stock sits and how fast it moves, but the verdict requires the item's role. The impeller's 274 days are the policy, not an accident: a critical spare with a sixteen-week replenishment lead time, whose absence can hold a customer's vessel, justifies a buffer that a locally replenished consumable never would. Cutting it to match the filter's 30 days would disarm exactly the protection the customers depend on. The review should instead watch the buffer's assumptions: if the engine family retires or the maker opens a nearer depot, the policy must be recomputed, or the buffer ages into the obsolete-stock flag.

Answer 1 applies a league-table rule the lesson explicitly dismantled; low turnover can be negligence or deliberate protection, and only the item's role distinguishes them. Answer 3 invents a false limitation: turnover is computable for any item with usage and stock records. Answer 4 reads both figures as equally healthy, ignoring that one policy is matched to its supply reality while the other is only assumed to be.

The takeaway: the measures flag; the item's role interprets.

Fact-check appendix (audit notes — not for entry)

Definitions of Inventory Optimisation, Service Level, Safety Stock, Cycle Stock, Reorder Point, Economic Order Quantity (EOQ), ABC Analysis, Inventory Holding Cost, Ordering Cost, Stockout, Fill Rate, Inventory Turnover, Replenishment Lead Time, and Inventory Position follow the lesson brief's key terms, defined at first use and used naturally in the blocks. — source: curriculum/lesson_briefs/module-13/lesson-13.3.md key terms.

Service level taught with its two common definitions, cycle service level as the target probability of no stockout and fill rate as the share of demand met from stock, with the organisation explicit about which is in use; higher targets need disproportionately more safety stock; no statistical safety-stock formula is presented. — source: research sheet logistics-scm-frameworks.md section 5; DO NOT CLAIM item 18.

Reorder point taught as expected demand during the lead time plus safety stock, carried over from Lesson 8.3's mechanics. — source: research sheet logistics-scm-frameworks.md section 5; curriculum/drafts/module-08/lesson-8.3.md.

EOQ taught only as principles: a classic model estimating an order quantity that minimises the sum of ordering and holding cost under strong simplifying assumptions (constant demand, full delivery at zero stock, fixed order cost, no quantity discounts or stockouts), treated as a model rather than a law; no formula is presented as a universal recipe. — source: research sheet logistics-scm-frameworks.md section 5 and DO NOT CLAIM item 18.

ABC analysis taught as a classification grouping items by relative importance, commonly annual consumption value, applied here to segment service levels and review intensity. — source: research sheet logistics-scm-frameworks.md section 5.

Inventory turnover taught as COGS divided by average inventory; days of supply as average inventory divided by average daily usage, approximately 365 divided by turnover when usage aligns with the cost figure; the lesson states the denominator in use. — source: research sheet logistics-scm-frameworks.md section 5.

Postponement (delayed differentiation) taught as keeping the product generic and deferring final configuration (dyeing, labelling, local packaging, final assembly) until real demand is known; suited to high demand uncertainty. — source: research sheet logistics-scm-frameworks.md section 8.

Network design trade-off taught symmetrically: more stocking points cut delivery time and outbound transport cost but add facility and inventory cost; centralised networks do the reverse. Pooling (risk pooling across stocking points) taught as a concept with expressly illustrative arithmetic; no square-root-law or other formula is claimed as universal; the correlated-demand limit is stated. — source: research sheet logistics-scm-frameworks.md section 8 (network design) and general supply chain knowledge.

Lead time components (order processing, preparation/pick-pack, transport and handling, customs clearance internationally) and OTIF as the strict on-time-and-in-full fulfilment measure, referenced as upcoming in Lesson 13.4. — source: research sheet logistics-scm-frameworks.md section 7.

Bullwhip causes (forecast updating/demand signal processing, order batching, price fluctuations, rationing/shortage gaming) referenced only as already taught in Lesson 13.1; no amplification magnitudes claimed. — source: research sheet logistics-scm-frameworks.md section 8; curriculum/drafts/module-13/lesson-13.1.md.

Holding-cost and failure-cost families (capital, space, insurance, shrinkage, obsolescence; lost margin, expediting, lost customers, contractual damage) are general business knowledge consistent with Lesson 10.1's cost families; no benchmark percentages or return figures are claimed anywhere in the lesson. — source: general business knowledge; curriculum/drafts/module-10/lesson-10.1.md.

All arithmetic (reorder points 580 and 930 from 330 units of expected lead-time demand plus 250 or 600 units of safety stock; expected stockout costs GHS 6,000, GHS 600, GHS 1,440, and GHS 144 from 1.2 or 0.12 expected events;

extra carrying costs GHS 2,100 and GHS 2,000; pooling release of 120 cartons, GHS 18,000 of working capital and GHS 900 a year; turnover of 12 and about 1.3; days of supply of about 30 and about 274 from daily usage of about 1,315 and about 164 dollars) is computed from expressly illustrative teaching figures labelled as such in each worked example; no real-company data is used. — source: WRITING_SPEC.md fact-safety rule 5.1.

Companies in examples are generic (a health-and-wellness distributor in Accra, a food wholesaler with stores in Tema and Kumasi, a marine spare-parts distributor in Rotterdam); no factual claims are attached to real named firms; the Ghana settings (Accra, Tema, Kumasi) and the international setting (Rotterdam) satisfy the example-placement rule. — source: WRITING_SPEC.md section 4.

Cross-references by name, no re-teaching: Lesson 5.4 Shipment Tracking and Customer Communication (communicating when promises slip); Lesson 7.1 Ocean Freight Operations and Vessel Schedules and Lesson 7.2 Air Cargo Operations and Cargo Acceptance (transit speeds on international lanes); Lesson 8.3 Inventory Control, FIFO, and FEFO (FIFO/FEFO, ABC analysis, reorder point, EOQ, the formulas-are-servants note); Lesson 8.4 Warehouse Documentation and Stock Reconciliation (record discipline); Lesson 10.1 Components of Total Logistics Cost (carrying cost, costs of failure, air-versus-sea total-cost reading); Lesson 13.1 Demand Forecasting and Supply Planning (forecast error, S&OP cycle, bullwhip, postponement from the planning side); Lesson 13.2 Transport Route and Network Optimisation (stocking-point trade-off); forward reference Lesson 13.4 Logistics Performance Indicators and KPIs (OTIF and fulfilment measures). — source: WRITING_SPEC.md section 7; lesson briefs for modules 5, 7, 8, 10, and 13.

Lesson 13.4 — Logistics Performance Indicators and KPIs

Summary

This lesson introduces performance measurement as a tool for monitoring and improving logistics operations. Learners will examine key performance indicators used in transportation, warehousing, inventory management, cost control, and customer service. The lesson also explains how performance data is collected, interpreted, and used to identify inefficiencies, establish targets, and support continuous improvement.

Learning objectives

- Define logistics performance measurement and key performance indicators.
- Explain why performance monitoring is important in logistics management.
- Identify common transportation, warehousing, and inventory KPIs.
- Describe how delivery reliability and order accuracy are measured.
- Explain the relationship between operational KPIs and customer satisfaction.
- Distinguish between performance targets and actual results.
- Calculate selected logistics KPIs using supplied operational data.
- Explain how dashboards and performance reports support management decisions.
- Identify limitations of poorly defined or misleading performance indicators.
- Recommend appropriate KPIs for a specified logistics operation.

Key terms

Key Performance Indicator (KPI) — A measurable indicator used to evaluate progress towards a defined operational or business objective.

Performance Measurement — The systematic collection and analysis of information used to evaluate operational results.

On-Time Delivery (OTD) — The proportion of deliveries completed within the specified delivery time requirements.

On-Time In-Full (OTIF) — The proportion of orders delivered within the agreed time and with the complete required quantity.

Order Accuracy — The degree to which fulfilled orders match customer requirements.

Inventory Accuracy — The extent to which recorded inventory information matches verified physical stock.

Warehouse Productivity — A measure of warehouse output relative to resources such as labour hours or equipment time.

Transport Utilisation — The proportion of available transportation capacity used during operations.

Cost per Shipment — The logistics cost allocated to an individual shipment under a defined costing method.

Order Cycle Time — The time between receipt of an order and completion of the defined fulfilment process.

Benchmarking — The comparison of performance against internal targets, historical results, or relevant external standards.

Performance Dashboard — A visual display presenting selected performance indicators and operational information.

Continuous Improvement — An ongoing approach to identifying and implementing changes that improve processes and performance.

Teaching content

KEY CONCEPT

From opinion to management: performance measurement and KPIs

Module 13 has built the plans: Lesson 13.1, Demand Forecasting and Supply Planning, produced the forecast; Lesson 13.2, Transport Route and Network Optimisation, shaped the routes; Lesson 13.3, Inventory Optimisation and Service Levels, set the stock policies. This lesson closes the module with the discipline that decides whether those plans actually work: performance measurement, the systematic collection and analysis of information used to evaluate operational results.

The central instrument of performance measurement is the key performance indicator, usually written KPI. A KPI is a measurable indicator used to evaluate progress towards a defined operational or business objective. Two parts of that definition matter. A KPI is measurable, meaning it is computed from operational records by a stated formula; and it is tied to an objective, meaning somebody decided in advance what level represents success. Many things in a warehouse can be counted, but a count becomes a KPI only when it has a formula, a target, a data source, and an owner.

Monitoring matters because logistics work is dispersed, time-critical, and judged by customers who do not care about internal explanations. Performance monitoring serves five purposes.

1. It makes service visible. Without measurement, a business learns about late deliveries when customers complain; with measurement, it knows its delivery reliability before the customer tells it.
2. It exposes inefficiency. Costs, idle capacity, and errors hide inside daily routine until a number names them.
3. It enables targets. A target set against measured baseline is a commitment; a target set against a feeling is a wish.
4. It supports continuous improvement, the ongoing approach of identifying and implementing changes that improve processes and performance. You cannot improve what you never measure.
5. It turns argument into evidence. Meetings about whether service is good enough end quickly when both sides read the same number.

In short, measurement converts logistics from opinion into management.

KEY CONCEPT

The behaviour rule: people optimise what you measure

A familiar saying in management holds that what gets measured gets managed. The deeper truth is sharper: people optimise what you measure. Staff and managers adapt their behaviour to the indicators placed before them, rationally, because measurement is how their work is judged. The design question is therefore not whether to measure, but whether the chosen measures reward the right behaviour, or whether they reward cleverly wrong behaviour.

Consider three measures steering somewhere their designers never intended.

1. A transport fleet measured only on cost per load will learn to wait. Holding departures until every truck is full cuts the cost per load beautifully, and quietly destroys delivery reliability, because orders now wait for the truck rather than the reverse.
2. A warehouse measured only on picking speed will learn to rush. Lines picked per hour climbs, and order accuracy slides, because speed measured alone is speed purchased with errors.
3. A distribution team measured only on on-time delivery will learn to split orders. Delivering whatever is ready on time protects the number while customers receive half-deliveries.

In each case no dishonesty is required, only a badly aimed measure. The lesson for the KPI designer is to define, for every indicator, the behaviour it is supposed to encourage, and to check that no reasonable person could improve the number while damaging the business. Where such a loophole exists, pair the measure with its counterweight: speed with accuracy, cost with service, availability with cost of carrying stock. The behaviour rule also explains why a few well-chosen indicators outperform a dashboard of everything, as the hierarchy discussion below develops.

KEY CONCEPT

Organising the KPI set: leading and lagging indicators, and the hierarchy

Performance indicators differ in when they speak. A lagging indicator reports a result after the fact — last month's OTIF, last quarter's claims ratio, the freight cost per unit actually paid — and is indispensable for accountability: it records the score honestly. A leading indicator predicts a result before it arrives, by measuring the conditions that produce it. Inventory accuracy leads service: records that disagree with the shelf today become tomorrow's short deliveries. Dock-to-stock time, the elapsed time from a vehicle's arrival at the receiving dock until goods are put away and pickable, leads dispatch reliability. Forecast error, measured by the MAPE taught in Lesson 13.1, Demand Forecasting and Supply Planning, leads availability; capacity utilisation trends lead cost and service pressure, because a fleet filling towards its limits will soon miss service or pay overtime and spot rates. A balanced set holds both kinds: lagging indicators tell management the score, leading indicators forecast the score while there is still time to change it — a review built only on lagging numbers is an inquest, a review built on both is a steering instrument.

The second organising principle is the hierarchy of three levels, each feeding the one above. Strategic KPIs are a small set, usually fewer than ten, tied to the service promise and the business plan — OTIF, overall logistics cost trend, inventory days of supply — reviewed monthly to answer one question: are we keeping the promises the strategy makes? Operational KPIs are process measures reviewed weekly or monthly by the managers who run the operation — order cycle time, freight cost per tonne-kilometre, inventory accuracy, vehicle capacity utilisation. Task-level KPIs are daily or per-shift measures used by supervisors and teams, such as lines picked per labour hour and orders dispatched per vehicle. The hierarchy works through cascade: lines picked per hour feeds order cycle time, and order cycle time feeds OTIF. A task-level number that cannot be traced upward to any operational or strategic objective is measuring habit rather than purpose, and should be challenged.

Together the two principles enforce the few-well-chosen rule: each level and each family carries only what its audience can act on — a handful at the top, more detail operationally, the most granular at task level. A single flat dashboard of forty indicators serves nobody; a measure must earn its place by changing a decision, otherwise it is decoration with a formula.

KEY CONCEPT

Service KPIs: measuring delivery reliability and order accuracy

Service KPIs measure whether the promise made to the customer was kept. Three indicators carry most of the weight.

1. On-Time Delivery

On-time delivery, OTD, is the proportion of deliveries completed within the specified delivery time requirements. The usual formula is: OTD percentage equals deliveries completed on time, divided by total deliveries, multiplied by 100. It measures the time half of the promise. Its weakness is that a delivery counted as on time may still be incomplete.

2. On-Time In-Full

On-time in-full, OTIF, is the proportion of orders delivered within the agreed time and with the complete required quantity. The formula is: OTIF percentage equals orders delivered both on time and in full, divided by total orders delivered, multiplied by 100. OTIF is deliberately strict: an order fails if it is late or short, and an early but incomplete delivery still fails. One number, two conditions, no averaging.

3. Order Accuracy

Order accuracy is the degree to which fulfilled orders match customer requirements: the right product in the right quantity, verified when the order is prepared and again at receipt. It isolates the picking and packing quality that OTIF blends with transport performance.

The composite service measure is the perfect order rate, which extends OTIF with two further conditions: the goods arrived undamaged, and the documentation and invoice were correct. Because one order must clear all conditions at once, some conventions compute it by multiplying the component success rates. The product is punishing. Illustrative figures: four components of 95, 95, 99 and 98 per cent multiply to roughly 87.6 per cent, although every component looks strong. This multiplicative fragility is the perfect order's whole point: it prices every failure into one number.

KEY CONCEPT

Speed KPIs: order cycle time and the parts of lead time

Service KPIs say whether the promise was kept; speed KPIs explain the promise itself and diagnose why it is missed.

The central measure is order cycle time, the time between receipt of an order and completion of the defined fulfilment process, from order entry through picking and dispatch to completed delivery. It is measured per order and reported as an average, but the average alone misleads. An illustrative operation reports a two-day average order cycle time; the worst orders took five. The average hides the customers who suffered. Report the average and the spread, the worst cases and how often they occur, because the tail of the distribution is where broken promises live.

Order cycle time is the customer-facing sum of a set of components that sit inside the wider idea of lead time, the elapsed time from order or replenishment trigger to receipt of goods. Its components are worth separating, because each has a different owner and a different cure.

1. Order processing: entry, verification, credit and confirmation.
2. Preparation: picking, packing, and staging for dispatch, governed by warehouse process discipline as taught in Lesson 8.2, Receiving, Put-Away, Picking, and Dispatch.
3. Transport: the transit leg, governed by the route and network decisions of Lesson 13.2.
4. Crossing borders: order processing, preparation, transit and customs clearance all count internationally, which is why imported lines carry structurally longer and more variable lead times.

The diagnostic power of the split is simple. When order cycle time deteriorates, the component view shows which stage absorbed the time, and corrective effort goes where the time went rather than where anecdote points.

KEY CONCEPT

Cost KPIs: freight cost, cost per shipment, and the cost of holding stock

Cost KPIs answer a different question from service KPIs: are we buying this service efficiently? Three families matter.

1. Freight cost measures

Freight cost per unit divides total freight spend by units shipped, such as per pallet delivered. Freight cost per tonne-kilometre divides spend by tonnes moved multiplied by kilometres travelled, the standard comparator between lanes and modes because it normalises for both weight and distance. An illustrative shipment of 10 tonnes carried 200 kilometres at a total cost of 1,000 GHS has a cost of 0.5 GHS per tonne-kilometre. Figures of this kind are illustrative for teaching purposes.

2. Cost per shipment

Cost per shipment is the logistics cost allocated to an individual shipment under a defined costing method. The closing phrase carries the discipline: the allocation rule must be stated and kept identical month after month, because a trend in which fuel was included one month and excluded the next measures the accountant's mood, not the operation.

3. Cost of carrying stock

Inventory turnover and days of supply, taught in Lesson 8.3, Inventory Control, FIFO, and FEFO, and applied in Lesson 13.3, belong in every logistics cost review because stock is bought logistics capacity. Restated briefly: inventory turnover equals cost of goods sold divided by average inventory, and days of supply equals average inventory divided by average daily usage; state which denominator you used. Falling turnover means money standing on shelves.

Above all three sits the cost-to-serve idea, which extends the total-cost lens of Lesson 10.1, Components of Total Logistics Cost, downwards to the individual customer or channel: the full cost of serving one customer, including order handling, picking, delivery frequency, and returns.

Cost-to-serve explains a truth that flat cost averages conceal: two customers buying the same revenue can cost very different amounts to serve.

KEY CONCEPT

Asset and quality KPIs: utilisation, accuracy, and damage

The third cluster of indicators watches the assets the business owns and the quality of the work done with them.

1. Capacity and transport utilisation

Transport utilisation is the proportion of available transportation capacity used during operations, and the same utilisation logic applies to containers and warehouse space: used capacity divided by available capacity, multiplied by 100. Utilisation is the core efficiency lever for both cost and emissions, and its most visible failure is empty running, travelling without paying cargo.

2. Warehouse productivity

Warehouse productivity is a measure of warehouse output relative to resources such as labour hours or equipment time, for example lines picked per labour hour. Its standing warning comes from the behaviour rule: productivity measured alone invites errors, so it is always read alongside accuracy.

3. Inventory accuracy

Inventory accuracy is the extent to which recorded inventory information matches verified physical stock, usually expressed as counted locations matching the record within tolerance, divided by locations counted, multiplied by 100. It is the output measure of the cycle counting taught in Lesson 8.3, resting on the record discipline of Lesson 8.4, Warehouse Documentation and Stock Reconciliation.

4. Dock-to-stock time

The elapsed time from arrival at the receiving dock to goods being put away and pickable. It is a leading indicator for dispatch reliability and a test of receiving discipline.

5. Claims and damage ratio

The value or number of damage and loss claims expressed as a share of shipments or of freight spend. It is a quality signal with a direct cost tail, and sensitive trades such as cold-chain food or pharmaceuticals watch it more closely than almost any other measure.

6. Forecast MAPE

The mean absolute percentage error taught in Lesson 13.1 measures planning quality, feeding the safety stock and service-level decisions of Lesson 13.3.

Quality measures are frequently the leading indicators of service measures: accuracy and damage today predict OTIF next month.

KEY CONCEPT

Targets and actuals: baseline first, then promise

A performance target is a defined level of a defined indicator, to be achieved by a defined date. An actual result is what measurement reports. The difference between the two is not an accounting nicety; it is the working agenda of every review meeting, because the gap is where management attention belongs. Confusing the two is a common failure: reporting the target as if it were the result confuses a plan with an achievement.

How are sound targets set? Three steps, and the order matters.

1. Baseline first
Measure before you manage. Establish what the indicator actually is today, computed by the agreed formula from honest records, for at least one full normal period. A baseline month of clean data is worth more than any borrowed number, because it is the only figure that describes this operation.
2. Target from your own trend and your own promise
Set the target just beyond the baseline, in the direction the service promise requires, with a date. Illustrative: OTIF measured at 89 per cent becomes a target of 92 per cent for next quarter, then 95 per cent the quarter after. The size of the step should reflect what the operation can resource, not what a conference chart once showed.
3. State the definition with the number
An indicator published without its formula and data source is not comparable across months or companies, so the definition travels with every target and every actual.

The comparison of performance against internal targets, historical results, or relevant external standards is called benchmarking, and it is a legitimate discipline when used carefully: your own targets, your own history, and the practices of others all inform. What benchmarking must never become is target-shopping, the importation of impressive numbers achieved by other businesses under other conditions. The next sections give that warning its full weight.

KEY CONCEPT

Reading KPIs together: tension, balance, and the road to customer satisfaction

No operational KPI should be read alone, because indicators push against one another. Freight cost per unit can fall because consolidation stretched delivery times. Warehouse productivity can rise because picking accuracy fell. Vehicle utilisation can climb because orders waited for full loads. Each of those pairs, cost against service, speed against accuracy, asset use against responsiveness, is a tension, and a management report that shows one side of a tension without the other is an invitation to celebrate a transfer as though it were a gain.

The reading rule is therefore: publish tension pairs together and judge them together. Freight cost per unit falling while OTIF also falls is not a win; it is a purchase of cost savings paid for in service, and the purchase may or may not be sensible, but it must be a decision, not an accident.

A small balanced set, one or two indicators from each family, service, speed, cost, asset, and quality, read side by side, will outperform a long list of solitary numbers every month.

The reason this discipline matters ultimately sits outside the company, with the customer. Operational KPIs and customer satisfaction are connected by a causal chain: inventory accuracy causes picking correctness; picking correctness and dock discipline cause OTIF; OTIF, order after order, becomes the customer's experience of reliability; experienced reliability becomes retention, and its absence becomes quiet defection that no single complaint ever announces. That is the practical meaning of the claim that logistics performance measurement protects revenue: it lets the business manage satisfaction before the survey instead of apologising after it. Surveys measure the end of the chain, after the fact; operational KPIs measure its causes, early enough to change the ending.

KEY CONCEPT

Dashboards, reports, and the operating rhythm

Measurement only pays when it is consumed on a rhythm: the dashboard, the report, and the meeting.

A performance dashboard is a visual display presenting selected performance indicators and operational information. A useful dashboard is small and disciplined: a handful of indicators, each shown against its target, each with a named owner, each displayed as a trend rather than a single snapshot, and flagged red, amber or green against thresholds agreed in advance. One page should be enough. A dashboard with forty tiles is not more informative than one with six; it is six informative signals drowned in thirty-four distractions.

The dashboard shows; the report explains. A monthly performance report adds commentary: why each red indicator is red, what the causes appear to be, what actions are under way, and what decisions management is asked to make: adding a vehicle, changing a carrier, raising safety stock. Dashboards and reports together support management decisions by replacing recollection and impression with a common, dated, defined set of facts.

Around the instruments runs the operating rhythm, the same five movements whether the operation is a single depot or a national network.

1. Define. Write down each KPI's formula, data source, owner, and review frequency.
2. Collect honestly. The records feeding the numbers follow the discipline of Lesson 8.4: posted at the time of the event, attributable, unadjusted without cause.
3. Review regularly. A weekly operational meeting for process indicators; a monthly management meeting for the strategic set.
4. Act. Every red indicator leaves the meeting with an owner, an action, and a date. A red number that leaves the meeting alone will be red again next week.
5. Audit the measure itself. Definitions drift, data quality decays, and behaviour adapts to the measure. Re-verify definitions and spot-check the data periodically.

This rhythm is continuous improvement made procedural, mirroring the classic plan, do, check, act cycle.

KEY CONCEPT

Choosing KPIs for an operation

The final skill of this lesson, and the capstone of the module, is recommending the right indicators for a specific operation. The method is short to state.

1. Start from the promise and the decisions
List what the operation has promised its customers, and the real management decisions it faces, carrier choice, delivery frequency, stock policy, staffing. KPIs exist to serve these; an indicator linked to neither a promise nor a decision is a candidate for deletion.
2. Cover the families
Build a starter set of roughly five indicators, one per family: service, such as OTIF; speed, such as order cycle time; cost, such as freight cost per unit or cost per shipment; asset use, such as vehicle capacity utilisation; and quality, such as inventory accuracy. A cold-chain food distributor would watch the claims and damage ratio closely from the start; a business living on imported lines would add a measure of clearance delays. Adjust the set to the operation, never to the template.
3. Define each completely
Formula, data source, owner, and review frequency, in writing, before the first number is produced.
4. Baseline before targets
Run the set for at least one normal period with no targets at all, then set trend targets from the baseline and the promise.
5. Review the set itself
Twice a year, ask which indicators changed a decision and which merely filled a page, and prune accordingly.

Two patterns fail repeatedly. The single-family dashboard, all cost or all service, steers the operation into the tensions described earlier. The exhaustive dashboard, every measure the systems can produce, dissolves attention. Five owned indicators, defined, baselined, and reviewed on a rhythm, will move an operation further than fifty unowned ones, and the worked examples that follow show it in practice.

WORKED EXAMPLE

An OTIF month at a packaged-foods distributor in Tema

A distributor of packaged foods based in Tema delivers to retail customers across Greater Accra. At the end of January, the logistics supervisor computes the month's service KPIs. All figures in this example are illustrative for teaching purposes.

Step 1: Gather the raw outcomes
During January the distributor completed 100 orders. Reviewing delivery notes and customer receipts, the supervisor records: 8 orders arrived later than the promised date; 5 orders were delivered short, meaning incomplete against the customer order; and 2 of the late orders were also short.

Step 2: Compute on-time delivery
Orders on time: 100 - 8 = 92. OTD = 92 ÷ 100 × 100 = 92 per cent.

Step 3: Compute the in-full rate
Orders delivered in full: 100 - 5 = 95. In-full rate = 95 per cent.

Step 4: Compute OTIF
OTIF counts only orders that were both on time and in full. Failures: 8 late + 5 short - 2 that failed both = 11 failed orders. OTIF = (100 - 11) ÷ 100 × 100 = 89 per cent. A tempting shortcut, averaging the 92 and the 95 per cent, gives 93.5 per cent, which is wrong, because it double-counts the 2 orders that failed on both counts.

Step 5: Split the failures diagnostically
The 89 per cent is one number hiding two diseases. Of the 8 late orders, 5 ran on the same regional lane in the same week, following the breakdown of a dedicated vehicle with no backup arranged; the other 3 carried imported lines that cleared late at the port, echoing the delay pressures examined in Lesson 9.4, Demurrage, Detention, and Storage Charges. All 5 short deliveries traced to one product family: the system showed stock available, but picking against inaccurate records produced short despatches, a process failure of the kind Lesson 8.2 addresses.

Step 6: Assign owners and actions
The transport supervisor owns a backup-vehicle arrangement for the dedicated lane. The warehouse supervisor owns a cycle count of the affected product family, with scan confirmation at the picking location. The procurement lead owns a safety-stock review on the late-clearing import line, applying the logic of Lesson 13.3.

What does this example demonstrate?
The headline OTIF of 89 per cent is a score, not a diagnosis. The split of late against short separates two different failures with different owners, different causes, and different cures.

Key lesson
Measure first, then dissect: an undivided service number tells you that something is wrong, and only the split tells you what to fix.

WORKED EXAMPLE

The consolidation that cost service: reading two KPIs together in Rotterdam

A food distributor based in Rotterdam serves supermarkets across the Netherlands from one distribution centre. All figures in this example are illustrative for teaching purposes.

Step 1: The baseline

At the start of the initiative the operation measures OTIF at 96 per cent and freight cost per pallet delivered at 20 euros. Supermarket orders arrive daily and travel mostly as part-load groupage shipments, the consolidation arrangement introduced in Lesson 5.3, Shipping Instructions and Cargo Consolidation.

Step 2: The cost initiative

The transport manager consolidates. Daily part-load deliveries are combined into two full-truck regional loops per week per region. Freight cost per pallet delivered falls to 16 euros, a 20 per cent reduction, and the initiative is reported to management as a clear success.

Step 3: The service response

Over the following two months, OTIF slips to 88 per cent. Average order cycle time lengthens, because orders placed after a loop has departed wait for the next one. Two supermarket customers complain formally about stock gaps on promoted lines, and one raises the deterioration at its supplier review.

Step 4: Read the two KPIs together

Neither number alone tells the story. The cost KPI shows a genuine 4-euro saving per pallet; the service KPI shows 8 points of OTIF paid for it. Together they show a transfer: the saving was manufactured by spending delivery reliability, the dynamic this lesson's tension pairs predict. Had the dashboard carried only freight cost, the initiative would have been declared a success until the customer losses, the true cost, arrived unmeasured.

Step 5: The rebalanced decision

Management redesigns rather than reverses. Planned replenishment orders, which supermarkets place on a known weekly cycle, are consolidated onto the loops under an agreed two-to-three-day promise, keeping most of the saving. Short-notice and promotion orders return to next-day direct delivery, accepting a higher cost per pallet on that defined slice of volume. The rebalanced targets: freight cost per pallet held near 18 euros on the blended mix, OTIF restored to 95 per cent or better.

What does this example demonstrate?

Two KPIs in tension must be read as a pair, and the correct response to an unbalanced trade-off is usually a redesign that keeps part of each benefit, not a retreat to either extreme.

Key lesson

A cost KPI without its service twin congratulates half a decision; publish them together or the operation will sell one to buy the other in secret.

WORKED EXAMPLE

A five-KPI rollout at a pharmaceutical distributor in Accra

A pharmaceutical distributor in Accra supplies pharmacies and clinics from one warehouse and a small fleet. A newly appointed distribution manager introduces performance measurement in one quarter. All figures in this example are illustrative for teaching purposes.

Step 1: Choose five KPIs, one per family

Service: OTIF, from delivery dates and completed quantities against orders. Speed: order cycle time, from order receipt to completed delivery. Cost: freight cost per shipment, monthly delivery cost divided by shipments, under a stated costing method. Asset: vehicle capacity utilisation, load volume over available volume. Quality: inventory accuracy from the monthly cycle counts. Each receives a written definition, a data source, a named owner, and a weekly computation with monthly review.

Step 2: Baseline one month, no targets

The first month produces the baseline: OTIF 91 per cent; order cycle time averaging 2.4 days with a worst case of 6 days; freight cost per shipment 680 GHS; vehicle utilisation 68 per cent; inventory accuracy 94 per cent. No targets are announced this month, so the numbers describe normal operations rather than staged behaviour.

Step 3: Set trend targets from the baseline and the promise

Targets for the next quarter: OTIF from 91 to 94 per cent; inventory accuracy from 94 to 98 per cent; vehicle utilisation from 68 to 75 per cent; order cycle time from 2.4 to 2.0 days with the worst case held below 3 days; freight cost per shipment held within 5 per cent of baseline while service improves, because forcing cost down during a service repair would recreate the tension trap this lesson warns against.

Step 4: Install the instruments

A one-page dashboard carries the five indicators, each against target with a twelve-week trend and red, amber or green flags. A thirty-minute review meeting runs every Monday morning: warehouse supervisor, transport supervisor, procurement lead, and the manager.

Step 5: Run the first reviews

In week two, inventory accuracy is flagged red at 93 per cent. Investigation traces the slippage to returned goods restocked without returns notes, breaching the documentation discipline of Lesson 8.4. Action: no return re-enters stock without a returns note; owner, the warehouse supervisor. In week three, OTIF is flagged red after an inbound shipment cleared late at the port. Action: raise safety stock on the five fastest-moving imported lines, applying Lesson 13.3's logic; owner, the procurement lead.

What does this example demonstrate?

Five defined, owned, baselined indicators on a fixed rhythm converted measurement into

management within a quarter; every red flag left its meeting with a name and an action attached.

Key lesson
A small, owned, well-defined set of KPIs beats a dashboard of everything; the review meeting, not the spreadsheet, is where performance is actually managed.

IMPORTANT NOTE

What gets measured badly gets gamed

The behaviour rule has a dark twin: what gets measured gets managed, and what gets measured badly gets gamed. Gaming is the adjustment of behaviour to flatter the number rather than to improve the business, and it requires no fraud, only pressure and a poorly designed measure.

Familiar patterns: Drivers learned that a delivery before the promised window counted against them, so trucks waited at the gate until the window opened, on-time statistics perfect, customers waiting at the gate with them. Warehouse teams recorded picking complete to satisfy a speed measure before the goods were physically staged. A branch redefined late from the promised delivery date to the dispatch date, and lateness vanished from the report while remaining untouched in the yard.

Two defences exist, and both are management work rather than technology.

1. Fix the definition in writing. A KPI's formula, data source, and counting rules are changed only deliberately, with the change dated and announced. Most gaming begins as definition drift, a small reinterpretation that makes the number kinder without anyone deciding anything.
2. Audit the measure itself. Periodically spot-check the data behind the indicator against physical reality: recheck a delivery date at the customer, recount a counted location. This is the discipline of Lesson 8.4, Warehouse Documentation and Stock Reconciliation, applied one level up: a KPI stands on records, and records that can be bent will be bent where the pressure to look good is greatest.

A measure that cannot survive an audit is not reporting performance; it is reporting the pressure its audience is under.

IMPORTANT NOTE

Borrowed benchmarks mislead: baseline and trend are your truth

The logistics world is full of impressive numbers: service levels that other companies report, cost ratios quoted at conferences, benchmark tables published by consultancies. On borrowed numbers this lesson's rule is blunt: there are no universal benchmark values for OTIF, inventory accuracy, utilisation, or any other KPI, and a text that quotes world-class percentages as targets is selling certainty it does not possess.

The reasons are structural. Two operations reporting the same KPI name may measure different things, because one counts an order on time when it arrives within the promised week and the other counts to the promised hour. They face different product mixes, different order profiles, different distances, different network shapes, and different promises made to different customers. A number produced by all of that history is not comparable to a number produced by different history, and setting your target at someone else's result commits your operation to a level whose conditions you cannot see and cannot copy.

Used correctly, external reference points still have a place. Benchmarking against published practice can tell you which indicators your industry considers worth measuring, and your own history, the internal benchmark, is genuinely comparable because it shares your definitions and your conditions.

The practical rule follows. Baseline first, target from your own trend and your own service promise, and treat every borrowed number as a curiosity until you have verified its definition. Baseline and trend are your truth; everything else is somebody else's story.

IMPORTANT NOTE

A KPI without an owner is decoration

The most common way measurement programmes die is not bad mathematics. It is orphaned indicators: numbers published, discussed, admired, and owned by nobody, which decay into wallpaper within a quarter.

An owner is a named person answerable for a specific indicator, with the authority to act on its causes and the duty to report on it at the review meeting. The definition matters in all three parts. Named, because a KPI belonging to the logistics department belongs to nobody. Answerable, because the owner is expected to explain the number, not merely produce it. Authorised to act, because holding a person responsible for a number whose causes they cannot touch is not accountability, it is scapegoating, and staff learn to defend themselves with explanations instead of improvements.

Three symptoms reveal an orphaned KPI set. First, review meetings that end without actions, because nobody present has the standing to commit to one. Second, indicators that stay red for months without consequence, which teaches the whole organisation that the dashboard is theatre. Third, a number whose definition nobody in the room can state when asked, which means the number can be reinterpreted by whoever finds it convenient.

The remedy costs nothing and is non-negotiable. No KPI enters the dashboard without a name beside it, and no red KPI leaves the review meeting without an action, a date, and the owner's agreement. Five owned indicators will outperform fifty orphaned ones, because measurement changes behaviour only at the point where a specific person decides that this number, on their watch, must move.

Practice questions (Lesson Checks)

Practice Question 1

A distributor in Kumasi delivered 80 orders during March. Reviewing the records, the logistics supervisor finds that 6 orders arrived later than the promised date, 4 orders were delivered short, and 2 of the late orders were also short.

Which figure is the distributor's correct OTIF for March?

1. 92.5 per cent, the proportion of orders that arrived on time.
2. 95 per cent, the proportion of orders delivered in full.
3. 90 per cent, the proportion of orders that were both on time and in full. ✓
4. 93.75 per cent, the average of the on-time rate and the in-full rate.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

OTIF counts only orders that clear both conditions at once: delivered within the agreed time and complete in quantity. The failures are the 6 late orders plus the 4 short orders minus the 2 counted in both lists, which gives 8 failed orders, leaving 72 of 80 orders both on time and in full. That is 72 ÷ 80 × 100 = 90 per cent.

Answer 1 is the on-time delivery rate, which is exactly the weakness OTIF exists to correct: an order counted on time may still be short. Answer 2 is the in-full rate alone, which ignores the time condition entirely. Answer 4 falls into the averaging trap from the Tema worked example: averaging 92.5 and 95 per cent double-counts the 2 orders that failed on both counts and flatters the true result.

The takeaway: OTIF is strict by design, late or short means failed, and overlapping failures must be subtracted once, not counted twice.

Practice Question 2

A distribution business consolidates its daily part-load deliveries into two full-truck regional loops per week. Freight cost per pallet falls by 20 per cent, but over the same two months OTIF slips from 96 to 88 per cent and two retail customers complain about stock gaps.

Which statement best describes how management should read these results?

1. The results confirm a successful initiative, because freight cost is the KPI the project targeted and it improved substantially.

2. The two results are unrelated, because cost performance and delivery reliability are measured by different departments from different data.

3. Service is the only KPI that matters, so the consolidation must be cancelled completely and every delivery returned to the daily schedule.

4. The results show a transfer from service to cost, and management must read the KPI pair together and rebalance, keeping part of the saving while restoring the service promise. ✓

Correct answer: Answer 4

Explanation

The correct answer is Answer 4.

Cost and service KPIs sit in tension: freight cost per unit can be cut by consolidating loads precisely because orders then wait for the truck, which is how OTIF falls. Read together, the two numbers describe a transfer, a genuine saving purchased with 8 points of delivery reliability, and the correct management response is the rebalanced decision shown in the Rotterdam worked example: consolidate planned replenishment orders that can carry a longer promise, keep direct delivery for short-notice orders, and set fresh targets for both indicators.

Answer 1 celebrates half a decision; a cost KPI without its service twin congratulates the transfer while the customer, the true cost, arrives unmeasured. Answer 2 is wrong because the indicators are causally linked, not merely adjacent on a dashboard. Answer 3 overcorrects to the other extreme; the rebalanced design kept most of the saving, and cancelling everything would surrender it unnecessarily.

The takeaway: no cost result should be reported without its service twin, because the operation will otherwise quietly sell one to buy the other.

Practice Question 3

A new distribution manager at a frozen-food distributor in Tema must choose the company's first set of logistics KPIs. Senior colleagues offer four different recommendations.

Which option best describes the soundest starting set?

1. A full dashboard of every indicator the systems can produce, so that management can later select the ones that turn out to matter.

2. Freight cost measures only, because cost is the one result management can directly control and the figures are already reliable.

3. Published world-class benchmark values for OTIF, inventory accuracy and utilisation, adopted as the company's targets from the first month.

4. About five indicators covering service, speed, cost, asset use and quality, such as OTIF, order cycle time, freight cost per shipment, vehicle utilisation and inventory accuracy, each with a formula, an owner, a baselining month, and targets set from the company's own trend. ✓

Correct answer: Answer 4

Explanation

The correct answer is Answer 4.

The lesson's recommendation method starts from the service promise, covers the KPI families so that no tension trades in the dark, defines each indicator completely, baselines before targeting, and sets targets from the company's own trend and promise. For a cold-chain operation the claims and damage ratio would soon join the set, but the five-family starter structure is the sound foundation, and the Accra pharmaceutical rollout showed it working end to end.

Answer 1 is the dashboard of everything, which dissolves finite attention across dozens of unowned numbers; nothing earns its place by changing a decision. Answer 2 is the single-family dashboard, which steers the operation into the cost-versus-service tension and invites exactly the consolidation trap of the Rotterdam example. Answer 3 imports borrowed benchmarks as targets, contradicting the fact-safety rule that no universal benchmark values exist: other businesses' results were produced under other definitions, mixes, distances and promises, and cannot be copied.

The takeaway: few, owned, well-defined indicators, baselined and trended, beat exhaustive, orphaned, or borrowed measurement every time.

Fact-check appendix (audit notes — not for entry)

Definitions of key performance indicator, performance measurement, on-time delivery, on-time in-full, order accuracy, inventory accuracy, warehouse productivity, transport utilisation, cost per shipment, order cycle time, benchmarking, performance dashboard and continuous improvement follow the lesson brief 13.4 key terms. — source: lesson brief 13.4, Key Terms (curriculum database).

OTIF percentage equals on-time-in-full deliveries divided by total deliveries, multiplied by 100; strict OTIF fails an order that is late or short, and an early but incomplete delivery still fails. — source: research sheet logistics-scm-frameworks.md, section 7 (OTIF, MetricHQ).

The perfect order rate extends OTIF with damage-free condition and correct documentation/invoicing, and some conventions compute it by multiplying component success rates. — source: research sheet logistics-scm-frameworks.md, section 7 (perfect order rate; medium confidence on exact components, taught with the some-conventions qualifier).

Order cycle time is measured from order receipt to delivery; dock-to-stock time is measured from arrival at the receiving dock to stock being pickable. — source: research sheet logistics-scm-frameworks.md, section 7.

Inventory accuracy is the share of counted locations or SKUs matching the system record within tolerance, the output measure of cycle counting. — source: research sheet logistics-scm-frameworks.md, section 7.

Freight cost per unit divides spend by units shipped, and freight cost per tonne-kilometre divides cost by tonnes moved per kilometre; it is the standard mode and lane comparator. — source: research sheet logistics-scm-frameworks.md, section 7.

Capacity utilisation is the share of available vehicle, container or warehouse capacity actually used; it is a core efficiency lever for cost and emissions, and empty running is travelling without paying cargo. — source: research sheet logistics-scm-frameworks.md, sections 7 and 6.

Claims and damage ratio expresses damage or loss claims as a share of shipments or freight spend. — source: research sheet logistics-scm-frameworks.md, section 7 (medium confidence on typical values; no values quoted in the lesson).

MAPE equals the average of the absolute difference between actual and forecast divided by the absolute actual, multiplied by 100, and is fragile near zero actuals. — source: research sheet logistics-scm-frameworks.md, section 7 (forecast accuracy).

Lead time runs from order or trigger to receipt, and its components include order processing, production or pick-pack, handling, transport and, internationally, customs clearance. — source: research sheet logistics-scm-frameworks.md, section 7.

Inventory turnover equals cost of goods sold divided by average inventory; days of supply equals average inventory divided by average daily usage, with the denominator stated. — source: research sheet logistics-scm-frameworks.md, section 5 (taught in Lessons 8.3 and 13.3; restated here only).

FTL dedicates a vehicle to one shipper while LTL and groupage combine several shippers' freight, cheaper per unit but with more handling; consolidation trades cost against transit and handling. — source: research sheet logistics-scm-frameworks.md, section 6.

PDCA, plan, do, check, act, is the Deming/Shewhart continuous improvement cycle. — source: research sheet logistics-scm-frameworks.md, section 13.

No universal OTIF benchmarks or specific retailer OTIF policies are claimed; benchmark figures are program-specific and time-bound, so the lesson teaches baseline-and-trend target setting. — source: research sheet logistics-scm-frameworks.md, DO NOT CLAIM item 9, and WRITING_SPEC.md rule 5.1.

Lead time is not just transport: order processing, preparation, transit and, across borders, customs all count; and cold-chain logistics is temperature-controlled storage and transport with continuous monitoring. — source: research sheet logistics-scm-frameworks.md, sections 7 and 1.

Leading versus lagging indicators, the strategic-to-task KPI hierarchy, the cost-to-serve lens, dashboard design practice (targets, owners, thresholds, trends), the define-collect-review-act-audit operating rhythm, and the sayings that what gets measured gets managed and that a KPI without an owner is decoration are general business knowledge, presented without attribution to any specific author. — source: general business knowledge.

All quantities, percentages, currency amounts, dates and company situations in the worked examples and inline illustrations (100 orders with 8 late, 5 short and 2 overlapping; 80 orders with 6 late, 4 short and 2 overlapping; 95/95/99/98 per cent perfect-order components; 20 to 16 euros per pallet; 680 GHS per shipment; 91/94/98 per cent and 68 to 75 per cent targets; 10 tonnes over 200 kilometres at 1,000 GHS) are illustrative for teaching purposes. — source: WRITING_SPEC.md rule 5.1.

Module 13 Assessment — 12 questions

Enter via the instructor portal's Configure assessment step. Green ticks mark correct answers — do not enter them; enter options as-is.

Question 1 (tests lesson 13.1)

1

A supplier of industrial consumables plans next quarter's purchases for two items. Item A is an established cleaning chemical with four years of steady monthly sales records. Item B is a new hand-tool range launching next month, never sold in this market. At the planning meeting, a colleague proposes applying the same simple quantitative method to both items, because a single company-wide method is easier to administer.

Which option best evaluates the proposal?

1. The proposal is sound, because a consistent company-wide method reduces forecast error on both items regardless of the data available for each.

2. The proposal is sound for Item B but not for Item A: new products need quantitative methods as soon as decisions loom, while established items respond best to expert judgement.

3. The proposal should be amended: quantitative forecasting fits Item A, whose long, steady history lets a method learn the pattern, while Item B, with no sales history, must be forecast qualitatively through expert judgement, structured market surveys, or a Delphi-style expert panel. ✓

4. The proposal should be amended in the opposite direction: expert judgement should carry Item A because experienced salespeople outperform formulas on mature products, and Item B should wait for six months of history before any forecast is attempted.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

The choice of forecasting method is a matter of fit between the method and the data. Quantitative forecasting learns from numerical history and assumes the pattern it finds continues into the future; Item A, with four years of steady monthly records, is exactly the case that rewards such methods, whether a moving average, exponential smoothing, or a causal model. Qualitative forecasting, built on expert judgement, structured market surveys, and Delphi-style panels that aggregate independent expert opinion in anonymous, iterated rounds, is the method of necessity where numbers do not exist. Item B has never sold in this market, so

structured judgement is the only honest basis for the launch commitments that lead times force now.

Answer 2 inverts the logic, assigning quantitative methods to the item without history and judgement to the one with data. Answer 1 elevates administrative consistency over fit; a consistent method that ignores each item's data situation produces a predictable failure on the new range. Answer 4 inverts the assignment again and adds the waiting fallacy: postponing the forecast until history exists does not remove the purchasing and space decisions, it only strips them of a stated demand assumption.

The takeaway: match the method to the data, and when there is no history, structure judgement rather than pretend history exists.

Question 2 (tests lesson 13.1)

2

A distributor records monthly demand for a fast-moving SKU as 300, 310, 320, 330, 340, and 350 cartons for months 1 to 6. All figures are illustrative for teaching purposes. The company forecasts with a three-month moving average, so the forecasts made for months 4, 5, and 6 were 310, 320, and 330 cartons, and each month's replenishment was planned from that forecast. The warehouse reports recurring stockouts.

Which option best describes the month 7 forecast and what the error record shows?

1. Forecast 340 cartons; every forecast so far has fallen exactly 20 cartons below the actual that followed, a one-directional pattern that is bias from the rising trend an averaging method lags, so the remedy is a trend-aware method or market intelligence, not a larger safety stock. ✓

2. Forecast 350 cartons; the most recent month is always the soundest estimate, and the moving average should be abandoned as unnecessarily complicated.

3. Forecast 340 cartons; the forecasts have tracked the series closely, so the method is accurate and the stockouts the warehouse reports must be unrelated to forecasting.

4. Forecast 325 cartons; the average of all six observed months is the most stable basis for the forecast, and the recurring misses are random noise that safety stock should buffer.

Correct answer: Answer 1

Explanation

The correct answer is Answer 1.

The month 7 forecast is the average of months 4 to 6: 330 plus 340 plus 350, divided by three, which is 340 cartons. The error record matters more than the number. The forecasts for months 4, 5, and 6 were 310, 320, and 330, and the actuals that followed were 330, 340, and 350: every forecast missed low, by the same 20 cartons. Errors scattered on both sides of a sound centre are noise, to be buffered with safety stock; errors clustered in one direction are bias, and this series shows textbook bias — demand is rising steadily, and an averaging method lags a trend by construction. Because each reorder point is built from the forecast of demand during the lead time, a persistently low forecast starts every replenishment cycle from the wrong base, which is exactly why stockouts recur even though the buffer was sized correctly for scatter.

Answer 2 abandons the method for the last observation, which still chases the trend one period behind and carries no error discipline. Answer 3 reads close tracking as accuracy and ignores the direction of the misses, the exact confusion that lets bias survive. Answer 4 averages all six months, a figure no moving-average policy produces, and mislabels systematic error as noise — the mistake that leads planners to buy ever-larger buffers against a model that could simply be corrected.

The takeaway: compute the forecast, then read the errors — fix bias in the method, and buffer only noise with stock.

Question 3 (tests lesson 13.1)

3

A manufacturer of cooking oil sells through one wholesaler, whose monthly orders have historically tracked consumer offtake closely. The manufacturer then runs a deep one-month promotional discount. During the promotion month the wholesaler's orders roughly triple; the following month they fall well below their normal level; retail data show that what consumers actually bought barely moved. The sales manager proposes treating the surge as evidence of booming demand and expanding production.

Which option best names what the order pattern actually shows?

1. The pattern is a shift in consumer demand that the retail data have not yet captured, so expanding production now positions the manufacturer ahead of the market.

2. The pattern is shortage gaming: the wholesaler inflated orders to secure a larger allocation during a shortage and cancelled once the scare passed.

3. The pattern is random variation in the wholesale tier, and no management response is required beyond watching the next few months.

4. The pattern is forward buying driven by the price fluctuation: the wholesaler stocked up at the promotional price and will live off that stock, so upstream has read a phantom surge

and slump; everyday stable pricing and sharing downstream sales data would flatten the swings. ✓

Correct answer: Answer 4

Explanation

The correct answer is Answer 4.

The lesson's bullwhip block names four causes of upstream order amplification: demand-signal distortion, order batching, price fluctuations, and rationing or shortage gaming. The scenario isolates one of them. The trigger was the promotional discount; the wholesaler bought cheap and stocked up, then lived off that stock the following month; and consumer offtake — the actual demand — barely moved. The surge and slump in orders are therefore forward buying: phantom demand created by the price calendar, not by consumers. Expanding production against it would build capacity and stock for a wobble the manufacturer's own promotion manufactured. The countermeasures that attack this cause are everyday stable pricing in place of deep promotions, and sharing downstream sales data so every tier forecasts from consumption rather than from distorted orders.

Answer 2 names a real cause but the wrong one: shortage gaming requires supply to run short and sellers to ration by order size, and none of that appears in the scenario. Answer 1 mistakes the distorted order signal for the market — the misreading that leads upstream tiers to invest in phantom demand. Answer 3 dismisses a patterned, explicable swing as random variation, forfeiting both the diagnosis and the remedy.

The takeaway: when upstream orders swing while end demand holds steady, find which bullwhip cause the pattern bears, and remove the cause rather than forecast its echoes.

Question 4 (tests lesson 13.2)

4

At a distribution company, the dispatch desk re-sequences delivery rounds every morning, and several of its best people spend their days rescuing problem deliveries: stock regularly expedited from one region to another to cover shortages, a consolidation hub used on one lane because it exists rather than because the flows justify it, and direct runs invented overnight for lanes the network was never designed to serve. The operations manager proposes a new incentive scheme to make the dispatch team's daily rescues faster.

Which option best assesses that response?

1. The response is appropriate: sharper incentives will make the daily rescues faster, which is what the operation currently needs most.

2. The response misreads the signal: recurring exceptions are network data — daily routing is compensating for a structure that no longer fits the flows — so the pattern should be escalated into a network review, the structure redesigned, and the routes then left to run quietly inside it. ✓

3. The response is unnecessary because route planning and network design are the same decision carried out at different times of day, so nothing needs to change beyond the daily schedule.

4. The response is premature: the operation should first buy a routing-software package, and only if the software cannot fix the daily patterns should the network be examined.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

The lesson draws a hard line between two decisions of different kinds. Route optimisation chooses the way a consignment travels within the network as it exists; network optimisation changes the structure itself — facilities, links, and flows — and is an investment taken rarely and deliberately. The symptoms listed are not routing failures. Stock repeatedly expedited between regions, a hub used because it exists rather than because the flows justify it, and direct runs invented overnight are the recognisable signature of daily routing being asked to compensate for a network that no longer fits the flows inside it. The lesson's rule is that recurring exceptions are network data: escalate the pattern into the network review, fix the structure, and let the routes run quietly inside it.

Answer 1 professionalises the firefighting, rewarding the rescue of deliveries instead of improving the system that manufactures the rescues — precisely how firefighting becomes institutionalised among an operation's best people. Answer 3 dissolves a distinction the lesson insists on: the two decisions differ in kind, in cost, and in reversibility, which is why they need different data, approvals, and review cycles. Answer 4 reaches for software first, but tools rank the options they are given; a routing package will optimise the runs inside a misfit structure faster, not repair the structure.

The takeaway: design the network first, then run routes within it — recurring daily exceptions are evidence about the network, not about the dispatch desk.

Question 5 (tests lesson 13.2)

5

A part-load carrier serves seven market towns. Between every pair of towns a small trickle of consignments moves each week; no single pair generates anything close to a full vehicle. A

newly appointed manager proposes abolishing the regional hub and running vehicles directly between every pair of towns, arguing that the hub adds distance, handling, and delay to every consignment.

Which option best evaluates the proposal?

1. The proposal is sound, because the shortest distance between two towns is always the cheapest way to move freight between them, whatever the volumes on the lane.

2. The proposal is sound for the thin flows but should exempt any pair of towns that generates a full vehicle of its own.

3. The proposal would multiply part-load journeys: with thin flows on every pair, direct lines would need far more connections and would run mostly empty, while the hub consolidates many small consignments into full outbound vehicles — the structure this flow pattern justifies, its price being distance, handling, and waiting at the hub. ✓

4. The proposal should be adopted for half the towns as a trial, because hub and direct structures are interchangeable where flows are thin.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

Structure follows flow. A hub earns its place where many origins and destinations exchange small consignments: it collects many part-loads, sorts them, and dispatches each outbound vehicle full. With seven towns and thin flows on every pair, direct lines would need one route per pair — seven towns give twenty-one direct connections against seven spokes — and each direct vehicle would run at a small fraction of capacity, with the cost and emissions of mostly empty journeys repeated on every leg. The hub's price is real: every inter-town consignment travels two legs, is handled at the hub, and waits for the outbound departure. But where no pair generates a full vehicle, that price is lower than the direct alternative of running largely empty vehicles — which is why part-load and groupage carriers typically operate hub-and-spoke terminal networks.

Answer 1 states the shortest-distance fallacy: distance per journey falls, but the number of journeys and the emptiness of each rise, and emptiness is what decides cost per unit carried. Answer 2 offers the hybrid inverted — the working pattern is direct lines where flows are dense and a hub where they are thin, never the reverse. Answer 4 treats the structures as interchangeable, when the whole skill is reading which flow pattern each structure serves.

The takeaway: count the flows, and give each lane the structure its volume supports.

Question 6 (tests lesson 13.2)

6

A manufacturer ships finished goods weekly from its plant to a retail depot in the north of the country, and its own shuttle vehicles return to the plant empty every week. A contract carrier offers to carry the northbound loads at a better rate than the internal cost of the shuttle, provided the manufacturer takes back its suppliers' returnable packaging — empty pallets and reusable crates — on the same vehicles from a collection point near the northern depot.

Which option best explains why the carrier can offer the better rate, and what the manufacturer should count?

1. The better rate is possible because cargo on the return leg turns a pure-cost journey into an earning one: the carrier's vehicles would otherwise run empty, burning fuel, driver time, and vehicle life with no revenue, and by supplying return cargo the manufacturer relieves exactly that empty-running problem — while the internal comparison should count the shuttle's own empty return in its cost per delivered load. ✓

2. The carrier is pricing aggressively to win the account and will raise the rate once the manufacturer is committed, so the offer should be refused whatever the return cargo.

3. The better rate reflects the carrier's newer vehicles, which burn less fuel per kilometre than the manufacturer's trucks, so the return cargo is incidental to the economics.

4. The return cargo benefits only the carrier, so the manufacturer should accept the rate but refuse to carry anything southbound.

Correct answer: Answer 1

Explanation

The correct answer is Answer 1.

Empty running — a vehicle travelling without paying cargo — is a major cost and emissions inefficiency: the empty leg burns fuel, driver time, and vehicle life at the full rate while earning nothing, and produces emissions with no freight to show for them. Flows are unbalanced in space, so return cargo is scarce on many lanes, which is exactly why carriers will price an outbound lane better where a shipper can offer traffic for the return direction. The manufacturer's pallets and crates flowing south turn the carrier's pure-cost journey into an earning one, and the relief of the carrier's empty-running problem is what funds the better northbound rate. The manufacturer should count honestly on its own side too: the internal shuttle's true cost per delivered load includes the empty return leg, so the internal figure the offer is measured against is flattered.

Answer 2 reads the offer as a loss-leader trick, but the economics stand on their own: filling a return leg genuinely releases cost that can be shared. Answer 3 attributes the rate to vehicles,

when the decisive line is cargo on the return leg. Answer 4 refuses the very cargo that sustains the saving, leaving the manufacturer's own packaging out of position and the carrier's leg empty again.

The takeaway: an empty kilometre is pure cost and pure emissions; structural backhauls make both sides of a lane cheaper.

Question 7 (tests lesson 13.3)

7

An importer stocks one critical spare part with an expected demand of 320 units during the replenishment lead time. All figures are illustrative for teaching purposes. At a 90 per cent cycle service level the safety stock is 80 units, giving a reorder point of 400. Raising the target to 95 per cent requires a safety stock of 140 units and a reorder point of 460, so 60 extra units are carried all year at an illustrative GHS 5 per unit per year. The item runs about 12 replenishment cycles a year; the raise halves expected failures from about 1.2 to about 0.6 a year, and a typical stockout costs an estimated GHS 800 in lost margin and expediting.

Which option best evaluates raising the target?

1. Raising the target costs nothing, because safety stock is insurance whose carrying cost is only an accounting entry rather than real money.

2. Raising the target is justified on the numbers: expected stockout cost falls from about GHS 960 to about GHS 480 a year, a saving of roughly GHS 480, against about GHS 300 of extra carrying cost for the 60 extra units — a gain of roughly GHS 180 a year before counting anything the estimate left out, and a gain that had to be bought with real carrying cost, because a service level is a policy choice with costs on both sides. ✓

3. Raising the target is unjustified, because the reorder point rises to 460 and any increase in the reorder point means over-stocking by definition.

4. The two targets deliver the same availability, because the reorder point already contains safety stock at both levels, so the choice can be made on convenience.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

A service level is a policy choice, and the choice has costs on both sides. At 90 per cent, 12 cycles a year carry a 10 per cent failure chance each, about 1.2 expected stockouts; at 95 per cent the expectation falls to about 0.6. At an estimated GHS 800 per event, expected stockout cost falls from about GHS 960 to about GHS 480 a year — a saving of roughly GHS 480. The raise is not

free: the reorder point moves from 400 to 460, the extra 60 units of safety stock are carried all year, and at GHS 5 per unit-year the additional carrying cost is about GHS 300. The comparison favours the raise by roughly GHS 180 a year before counting any stockout consequence the estimate left out — and the protection had to be purchased with genuine carrying cost, which is the whole discipline of setting the target deliberately.

Answer 1 treats carrying cost as an accounting fiction, but holding cost is real money in capital, space, insurance, and obsolescence whether or not an invoice names it. Answer 3 declares any higher reorder point over-stocking by definition, the league-table reflex; the question is whether the extra protection is worth its price, and here it is. Answer 4 denies that the targets differ, but the two reorder points are designed to buy different failure probabilities.

The takeaway: set the target deliberately, price both sides honestly, and let the comparison decide.

Question 8 (tests lesson 13.3)

8

A stationery wholesaler stocks some 2,000 items at four regional stores. A consolidation study splits the assortment into slow-moving specialist lines whose demand is uncertain and moves independently in each region, and fast-moving everyday lines that nearby schools and offices reorder within hours of running out. Management decides to pool the safety stock of the whole assortment into one central store, because pooling one product family in a trial had reduced total stock.

Which option best evaluates the decision?

1. The decision is sound and should be applied to the whole assortment as quickly as possible, because the more items are pooled, the larger the saving on every item.

2. The decision is unsound in principle, because pooling never reduces total stock, and the earlier trial's saving must have been measured incorrectly.

3. The decision should depend on unit value alone: the most expensive items should stay regional, and everything cheap should be pooled.

4. The decision overreaches: pooling pays best on the slow-moving lines whose regional swings partly cancel, while fast movers and items whose absence stops a customer's work within hours must live close to demand regardless, so the assortment should be pooled item by item along lines of importance and response need, as ABC-style classification supports. ✓

Correct answer: Answer 4

Explanation

The correct answer is Answer 4.

Pooling is real: when the same item is held at several sites, each site hedges its own local swings, and where those swings are independent they partly cancel under one pool, so the same availability can be designed with a smaller total buffer. But the lever has honest limits that a blanket decision ignores. The gains shrink when demands rise and fall together, and pooling must be paid for in response time to the periphery. The lesson's rule is to pool item by item: slow-moving lines with uncertain, independent regional demand are the natural candidates, while fast movers and items whose absence stops a customer's work within hours must sit close to demand regardless — exactly the split the study found in this assortment. ABC-style classification, grouping items by relative importance and pairing each class with the treatment it justifies, is the systematic instrument for drawing that line instead of leaving it to habit.

Answer 1 generalises from one successful family trial to the whole assortment, carrying pooling past the conditions that make it pay. Answer 2 denies the effect the trial demonstrated. Answer 3 substitutes unit value for the criteria that actually decide — variability, correlation, and response need — so the most expensive fast mover and the cheapest critical slow mover would both be misfiled.

The takeaway: pooling buys availability with a smaller total hedge, pays in delivery speed, and is decided item by item.

Question 9 (tests lesson 13.3)

9

A service company imports one critical component through an ocean pipeline of about six weeks — illustrative figures for teaching purposes — and holds a large safety stock against six weeks of demand variability and occasional port delays. A freight forwarder offers an air routing that delivers in about five days at a higher cost per consignment. The finance manager objects that the air rate is more than double the ocean rate per shipment and should be refused.

Which option best evaluates the objection?

1. The objection is correct, because freight cost per consignment is the measure that should decide between competing routings.

2. The objection is correct, because switching to air would force the company to hold more safety stock, since air shipments are smaller than ocean consignments.

3. The objection compares only one line of the total: the shorter, steadier pipeline shrinks the exposure window the safety stock exists to cover, so the same service can be kept with a much smaller buffer, and the mode choice should be made on transport and inventory cost together, as the total-cost lens requires. ✓

4. The objection is wrong because safety stock depends only on the item's unit cost, not on how the item is shipped.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

Safety stock is sized against what can go wrong during the replenishment lead time — the exposure window. A six-week ocean pipeline exposes the operation to six weeks of demand variability plus occasional port delay; a five-day air routing shrinks the window to days and, typically, steadies it. The same service promise can therefore be kept with a much smaller buffer, and the released stock is a genuine inventory saving. This is the total-cost discipline: freight savings and inventory savings live in different ledgers, so a mode that looks dearer per consignment can be cheaper to the business once the stock it eliminates is priced — and the decision must compare transport and inventory together rather than one line at a time.

Answer 1 lets a single line, cost per consignment, decide a system question — precisely the silo comparison the total-cost lens exists to correct. Answer 2 inverts the mechanism: the shorter pipeline reduces the exposure, and with it the buffer, not the reverse. Answer 4 denies the link between lead time and safety stock altogether; the buffer exists because demand and supply vary during the wait, so the length and the reliability of the wait are what set its size.

The takeaway: shrink the exposure and the buffer shrinks with it — judge the mode on transport and inventory cost together, never on the freight line alone.

Question 10 (tests lesson 13.4)

10

A distributor of household goods completed 150 orders in April. Checking delivery notes against customer orders — all figures illustrative for teaching purposes — the supervisor finds that 12 orders arrived after the promised date, 9 orders were delivered short of the ordered quantity, and 3 of the late orders were also short.

Which figure is the distributor's correct OTIF for April?

1. 92 per cent, the proportion of orders that arrived on time.

2. 94 per cent, the proportion of orders delivered in full.

3. 93 per cent, the average of the on-time rate and the in-full rate.

4. 88 per cent, the proportion of orders that were both on time and in full. ✓

Correct answer: Answer 4

Explanation

The correct answer is Answer 4.

OTIF counts only orders that clear both conditions at once: delivered within the agreed time and complete in quantity. The failures are the 12 late orders plus the 9 short orders minus the 3 that appear in both lists, which is 18 failed orders, leaving 132 of 150 orders delivered both on time and in full. That is 132 divided by 150, multiplied by 100 — 88 per cent.

Answer 1 is the on-time delivery rate, 138 of 150, which is exactly the weakness OTIF exists to correct: an order counted on time may still be short. Answer 2 is the in-full rate alone, 141 of 150, which ignores the time condition entirely. Answer 3 falls into the averaging trap: averaging 92 and 94 per cent gives 93 per cent, but the average double-counts the 3 orders that failed on both counts and flatters the true result. OTIF is deliberately strict — an order that is late or short has failed — and overlapping failures are subtracted once, never twice.

The takeaway: compute OTIF from the overlap-corrected failure count, then split the failures diagnostically, because the single number hides two different diseases — lateness and shortage — with different causes and different owners.

Question 11 (tests lesson 13.4)

11

A distribution centre introduces picking-speed targets. Lines picked per labour hour rise sharply, orders still leave on time, and the centre's manager reports the productivity gain to head office as a clear improvement. Over the same weeks, customer complaints about wrong items in delivered orders increase, and returns caused by picking errors climb.

Which option best describes how head office should read the report?

1. The report is reliable, because a rise in lines picked per labour hour is direct evidence that the picking process has improved.

2. The report shows a tension transfer: productivity measured alone invites errors, and speed purchased with accuracy damage is not an improvement, so the productivity measure must be published with its counterweight — order accuracy — and the two read together before any gain is declared. ✓

3. The report shows that the complaints are coincidental, because picking speed and picking accuracy are independent measures produced by different processes.

4. The report shows that the targets have failed and should be scrapped, with the centre returning to whatever picking pace it used before.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

People optimise what they are measured on, and productivity measured alone is one of the standard traps: lines picked per labour hour can be raised by rushing, and speed purchased with errors is not improvement. The behaviour rule requires the designer to state what behaviour each measure should encourage and to pair every measure with its counterweight — speed with accuracy, cost with service. Head office should therefore read the productivity number beside order accuracy, and the rising complaints and picking-error returns are the accuracy twin already speaking: part of the reported gain is a transfer from quality to speed, and the pair must be judged together before any improvement is declared.

Answer 1 celebrates a single-family reading, exactly how a tension transfer gets reported as victory. Answer 3 asserts independence, but the lesson's causal chain runs the other way: picking correctness feeds the customer's experience of reliability, so speed and accuracy are linked by construction, not coincidence. Answer 4 overcorrects; the remedy is not abandoning speed targets but defining them alongside their counterweight, so that improvements are real rather than purchased with defects.

The takeaway: publish tension pairs together and judge them together, or the operation will quietly sell one to buy the other.

Question 12 (tests lesson 13.4)

12

A newly appointed logistics manager finds that her company has never measured its delivery reliability. At a trade conference she hears that leading distributors report on-time in-full results of 99 per cent, and she announces to customers that the company will match 99 per cent from the following month.

Which option best assesses the announcement?

1. The announcement promises a number the operation has never measured, produced under definitions and conditions it cannot see: no universal benchmark values exist for OTIF or any other KPI, so the sound sequence is to define the measure, baseline it for at least one full normal period, and set the target from the company's own trend and its own service promise. ✓

2. The announcement is sound practice, because published figures from leading distributors are the most reliable available targets, and adopting them is what benchmarking is for.

3. The announcement is sound provided the manager raises the promise to 100 per cent, which removes any risk of disappointing customers.

4. The announcement should be implemented by reporting 99 per cent as achieved from the first month, since the target now defines the result the company is committed to.

Correct answer: Answer 1

Explanation

The correct answer is Answer 1.

There are no universal benchmark values for OTIF, inventory accuracy, utilisation, or any other KPI. A number reported by another operation was produced under its definitions, product mix, order profile, distances, network shape, and promises — conditions this company cannot see and cannot copy, starting with how tightly on time was even counted. The lesson's sequence is explicit: define the measure completely, baseline it for at least one full normal period, then set the target just beyond the baseline, in the direction the service promise requires, with a date. A target set against a measured baseline is a commitment; one set against a conference figure is a wish announced to customers. Benchmarking has a legitimate use — learning which indicators an industry considers worth measuring — and the internal benchmark, the company's own history, is the comparison that shares its definitions and conditions.

Answer 2 mistakes target-shopping for benchmarking, which is exactly the importation of impressive numbers the lesson prohibits. Answer 3 doubles the error and collides with the inventory lesson's finding that 100 per cent is not a strategy but a refusal to choose a target. Answer 4 reports the target as the result, confusing a plan with an achievement.

The takeaway: baseline first, then target from your own trend and your own promise — baseline and trend are your truth; everything else is somebody else's story.